FINITE GENERATION FOR VALUATIONS COMPUTING WEIGHTED STABILITY THRESHOLDS

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Abstract. We prove that if a weighted stability threshold of a log Fano pair is at most one, then any valuation computing it has a finitely generated associated graded ring. Thus we answer Question 5.3 in the paper of Blum, Liu, Xu, and Zhuang on the existence of the Kähler–Ricci soliton degeneration.

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1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

Blum, Liu, Xu, and Zhuang [BLXZ23] introduced weighted versions of the stability threshold of a log Fano pair in their proof of the existence of the Kähler–Ricci soliton degeneration. Let (X, Δ) be a log Fano pair (Definition 2.2), let r be a positive real number such that $L = -r(K_X + \Delta)$ is a Cartier divisor, and let $R = \bigoplus_{m\in\mathbb{N}} H^0(X, \mathcal{O}_X(mL))$ [BLXZ23, Section 2.1]. Fix a torus $T \subset \mathrm{Aut}(X, \Delta)$, where $T = \{1\}$ is allowed, a continuous function $g : \mathbb{R} \to \mathbb{R}_{>0}$, and a quasi-monomial valuation $v_0 \in \mathrm{Val}_X^{T,\circ}$ (Definitions 2.4 and 2.6), that is, v_0 is a non-trivial T-invariant quasi-monomial valuation on X whose log discrepancy $A_{X,\Delta}(v_0)$ is finite.

Let $v \in \mathrm{Val}_X^{T,\circ}$. For a positive integer m with $H^0(X, \mathcal{O}_X(mL)) \neq 0$, let $(s_1, \dots, s_N)$ be a basis of $H^0(X, \mathcal{O}_X(mL))$ compatible with both v_0 and v, that is, for any real number λ, each of the subspaces $\{s : v_0(s) \geq \lambda\}$ and $\{s : v(s) \geq \lambda\}$ is spanned by some of the s_i. Such a basis exists by [AZ22, Lemma 3.1]. Blum, Liu, Xu, and Zhuang defined the weighted expected vanishing order

$$S_g(v_0; v) = \lim_{m\to+\infty} \frac{\sum_{i=1}^N g\left(\frac{v_0(s_i)}{mr}\right) v(s_i)}{mr \sum_{i=1}^N g\left(\frac{v_0(s_i)}{mr}\right)},$$

where N and $(s_1, \dots, s_N)$ depend on m [BLXZ23, Section 4.1]. Equivalently, $S_g(v_0; v)$ is the integral of $g(x)y$ divided by the integral of $g(x)$ with respect to the compatible Duistermaat–Heckman measure of v_0 and v on $\mathbb{R}^2$ [BLXZ23, Theorem 3.3]. The weighted stability threshold is

$$\delta_{g,T}(X, \Delta, v_0) = \inf_{v\in\mathrm{Val}_X^{T,\circ}} \frac{A_{X,\Delta}(v)}{S_g(v_0; v)},$$

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and a valuation $v \in \mathrm{Val}_X^{T,\circ}$ computes $\delta_{g,T}(X,\Delta,v_0)$ if the infimum is attained at v (Definition-Proposition 2.15; see also [RTZ21, Section 6]). By [BLXZ23, Corollary 4.13], $\delta_{g,T}(X,\Delta,v_0) \geq 1$ if and only if (X,Δ,v_0) is T-equivariantly g-Ding semistable [BLXZ23, Definition 4.6], which generalizes the valuative criterion of Fujita and Li [Fuj19, Theorem 1.4], [Li17, Theorem 3.7].

Blum, Liu, Xu, and Zhuang asked whether, if $\delta_{g,T}(X,\Delta,v_0) \leq 1$ and $v \in \mathrm{Val}_X^{T,\circ}$ computes $\delta_{g,T}(X,\Delta,v_0)$, the associated graded ring $\mathrm{gr}_v R$ (Definition 2.18) is finitely generated [BLXZ23, Question 5.3]. Two choices of g and v_0 motivate the question. In the first, (X,Δ) is not K-semistable, $g(x) = e^{-x}$, and v_0 is the valuation minimizing the $\tilde{\beta}$-functional of Han and Li [BLXZ23, Theorem 1.2], which is non-trivial in this case, and the finite generation of $\mathrm{gr}_{v_0} R$ was conjectured by Han and Li [HL24, Conjecture 4.11]. In the second, v_0 is the valuation determined by a non-zero vector $\xi \in \mathrm{Hom}(\mathbb{G}_m, T) \otimes_{\mathbb{Z}} \mathbb{R}$, which is the setting of Kähler–Ricci g-solitons: Han and Li proved that a triple (X,Δ,ξ) admits a Kähler–Ricci g-soliton if and only if it is reduced uniformly g-Ding stable [HL23, Theorem 1.7] (see [BLXZ23, Theorem 4.9]).

Before this paper, [BLXZ23, Question 5.3] was answered in the following cases.

(1) When g is constant and $T = \{1\}$, for R defined with a positive integer r such that $r(K_X + \Delta)$ is Cartier [LXZ22, Theorem 1.1].
(2) When v is quasi-monomial and, for some vector $\xi \in \mathrm{Hom}(\mathbb{G}_m, T) \otimes_{\mathbb{Z}} \mathbb{R}$, either v is the twist of v_0 by ξ or v_0 is the valuation determined by ξ [BLXZ23, Theorem 5.4].

We first explain case (1). When g is constant, $S_g(v_0; v)$ depends only on v: by [BLXZ23, Proposition 3.6(1)], it is equal to the expected vanishing order of v in the definition of the stability threshold $\delta(X,\Delta)$ [LXZ22, Definition 2.8] (the pushforward of the compatible Duistermaat–Heckman measure of v_0 and v under the second projection is the Duistermaat–Heckman measure of v, whose barycenter is the expected vanishing order of v [BLXZ23, Section 2.2.1]). Thus $\delta_{g,\{1\}}(X,\Delta,v_0) = \delta(X,\Delta)$ for any v_0, and the valuations computing $\delta_{g,\{1\}}(X,\Delta,v_0)$ are the valuations computing $\delta(X,\Delta)$. In this case [LXZ22, Theorem 1.1] proves the Higher Rank Finite Generation Conjecture [Xu21, Conjecture 1.2] in the following form: if $\dim X = d$, r is a positive integer such that $r(K_X + \Delta)$ is Cartier, $R = \bigoplus_{m \in \mathbb{N}} H^0(X, -mr(K_X + \Delta))$, and $\delta(X,\Delta) < (d+1)/d$, then $\mathrm{gr}_v R$ is finitely generated for any valuation v computing $\delta(X,\Delta)$.

The main ingredient is special complements. Assume that $\dim X = d$, $\delta(X,\Delta) < (d+1)/d$, and v computes $\delta(X,\Delta)$. By [LXZ22, Lemma 3.1], for any $\mathbb{Q}$-divisor $D \geq 0$ with $D \sim_{\mathbb{Q}} -(K_X + \Delta)$, there exists a $\mathbb{Q}$-complement $(X, \Delta + \Gamma)$ of (X,Δ) with $\Gamma \geq \alpha D$ and with v an lc place of $(X, \Delta + \Gamma)$, where α is a positive rational number depending only on d and $\delta(X,\Delta)$. Since D is arbitrary, it can be taken general, and [LXZ22, Corollary 3.4] shows that there exists a special $\mathbb{Q}$-complement $(X, \Delta + \Gamma)$ of (X,Δ) in the sense of [LXZ22, Definition 3.3] with $v/A_{X,\Delta}(v)$ a monomial lc place.

By [LXZ22, Theorem 4.2], for a quasi-monomial valuation v, the ring $\mathrm{gr}_v R$ is finitely generated and the induced degeneration of (X,Δ) is klt if and only if v is a monomial lc place of a special $\mathbb{Q}$-complement. Here the valuation v is quasi-monomial: by [BJ20, Proposition 4.8(ii)] (the result is formulated in [BJ20] for klt varieties without boundary, and as noted in the introduction of [BJ20], the arguments there apply to klt pairs), v is the unique valuation, up to scaling, computing the log canonical threshold of the graded sequence of valuation ideals of v, and by [Xu20, Theorem 1.1], the log canonical threshold of a graded sequence of ideals on a klt pair, when finite, is computed by a quasi-monomial valuation (here the threshold is finite: it equals $A_{X,\Delta}(v) < +\infty$ because $v(\mathfrak{a}_\bullet(v)) = 1$ and v computes it [BJ20, Section 1.3 and Proposition 4.8(i)]). See also [KL26, Theorem 1.1] for the existence of quasi-monomial valuations computing $\delta(X,\Delta)$.

We refer the reader to [LXZ22, Theorems 1.2, 1.3, and 1.6] for the applications of [LXZ22, Theorem 1.1] to optimal destabilization, to the properness and projectivity of K-moduli spaces, and to the Yau–Tian–Donaldson conjecture for log Fano pairs. See also [BLZ22] and [XZ20].

Case (2) is enough for the applications in [BLXZ23]. If (X,Δ) is not K-semistable, $g(x) = e^{-x}$, and v_0 is the (non-trivial) valuation minimizing the $\tilde{\beta}$-functional of Han and Li, then [BLXZ23, Corollary 5.2] shows that $\delta_{g,\{1\}}(X,\Delta,v_0) = 1$ and that v_0 computes $\delta_{g,\{1\}}(X,\Delta,v_0)$,

and [BLXZ23, Corollary 5.7] deduces that $\mathrm{gr}_{v_0} R$ is finitely generated, which leads to [BLXZ23, Theorem 1.2]. When v_0 is determined by a non-zero vector $\xi \in \mathrm{Hom}(\mathbb{G}_m, T) \otimes_{\mathbb{Z}} \mathbb{R}$, [BLXZ23, Corollary 5.8] shows that $\mathrm{gr}_v R$ is finitely generated for any quasi-monomial valuation $v \in \mathrm{Val}_X^{T,\circ}$ computing $\delta_{g,T}(X, \Delta, v_0)$. As a consequence, any g-Ding polystable triple (X, Δ, ξ) [BLXZ23, Definition 4.10] is reduced uniformly g-Ding stable, and hence admits a Kähler–Ricci g-soliton [BLXZ23, Corollary 5.9]. See also [BLXZ23, Theorem 1.3] for the Yau–Tian–Donaldson conjecture for Kähler–Ricci solitons.

Both [BLXZ23, Theorem 5.4] and [BLXZ23, Corollary 5.8] assume that the computing valuation v is quasi-monomial. The result [BJ20, Proposition 4.8(ii)] used in case (1) is a statement about valuations computing $\delta(X, \Delta)$, and it does not directly apply to $\delta_{g,T}(X, \Delta, v_0)$: the expected vanishing order is replaced by $S_g(v_0; \cdot)$, and the infimum is taken only over T-invariant valuations.

We now explain the role of the additional assumptions in case (2). Let m be a positive integer and $(s_1, \dots, s_N)$ a basis of $H^0(X, \mathcal{O}_X(mL))$ compatible with v_0. The g-weighted (m, v_0)-basis type divisor of $(s_1, \dots, s_N)$ is the $\mathbb{R}$-linear combination of the divisors $\mathrm{div}(s_i)$ with coefficients proportional to $g(v_0(s_i)/(mr))$, normalized so that it is $\mathbb{R}$-linearly equivalent to $-(K_X + \Delta)$ [BLXZ23, Definition 4.1] (see Definition-Proposition 2.15). The weight of s_i is read from $v_0(s_i)$, so the definition uses the compatibility with v_0. When the basis is also compatible with v, the value along v of the g-weighted (m, v_0)-basis type divisor is the degree-m approximation of $S_g(v_0; v)$, that is, the fraction in the limit defining $S_g(v_0; v)$.

To obtain a special $\mathbb{Q}$-complement, one wants such a basis type divisor to contain a fixed positive multiple of an auxiliary $\mathbb{Q}$-divisor $G \geq 0$ with $G \sim_{\mathbb{Q}} -(K_X + \Delta)$. By [BLXZ23, Lemma 5.6], there exists a constant $c > 0$ such that, for any sufficiently large m, any g-weighted (m, v_0)-basis type divisor D compatible with G satisfies $D \geq cG$. Thus one wants a single basis compatible with all three of v_0, v, and G. Any two of the compatibility conditions with v_0, v, and G can be satisfied by one basis [AZ22, Lemma 3.1], but in general no basis satisfies three such conditions.

For instance, let p_0, p_1, and p_2 be distinct points of $\mathbb{P}^1$ and m a positive integer. For any i, a basis of $H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(m))$ compatible with the vanishing order at p_i contains exactly one section s with $\mathrm{mult}_{p_i} \mathrm{div}(s) = j$ for any $j \in \{0, 1, \dots, m\}$. Thus, for a basis $(s_0, \dots, s_m)$ compatible with the vanishing orders at p_0, p_1, and p_2, we would have

$$\sum_{k=0}^{m} \sum_{i=0}^{2} \mathrm{mult}_{p_i} \mathrm{div}(s_k) = \frac{3m(m+1)}{2} > m(m+1),$$

which contradicts the inequalities

$$\sum_{i=0}^{2} \mathrm{mult}_{p_i} \mathrm{div}(s_k) \leq \deg \mathrm{div}(s_k) = m, \qquad k = 0, \dots, m.$$

When g is constant and $T = \{1\}$, the weights do not depend on v_0, so one can work with the basis type divisors of [LXZ22, Definition 2.6] defined by arbitrary bases. Then only the compatibility conditions with v and with G remain, and [AZ22, Lemma 3.1] provides a basis satisfying both. Under either of the assumptions in case (2), in any sufficiently divisible degree, a basis consisting of T-eigensections that is compatible with v is also compatible with v_0, and the difficulty with three compatibility conditions does not occur.

Finite generation has also been studied for other valuations and in other settings. Xu and Zhuang proved finite generation for the minimizer of the normalized volume of a klt singularity [XZ25, Theorem 1.1]. See also [Che25] and the survey of Xu and Zhuang [XZ26, Theorem 39]. Han, Miao, Qi, Wang, and Zhang [HMQ+26, Theorem 1.1] extended the soliton degeneration to log Fano fibration germs, and their result specializes to [BLXZ23, Theorem 1.2] when the base is a point. L. Wang [Wan25] introduced the $\mathbf{H}^g$-invariant and constructed g-optimal degenerations. For log Fano cone singularities, D. Kim [Kim26, Theorem 1.1] proved that the δ-invariant weighted

by a T-invariant valuation is attained. Properness for the moduli of weighted K-semistable log Fano pairs is announced in [BLXZ25, v1, Section 1.5].

The goal of this paper is to answer [BLXZ23, Question 5.3] affirmatively.

We have the following theorem.

Theorem 1.1. *Let (X,Δ) be a log Fano pair and r a positive real number such that $L := -r(K_X+\Delta)$ is Cartier. Let $R := \bigoplus_{m\in\mathbb{N}} H^0(X,\mathcal{O}_X(mL))$, $T \subset \mathrm{Aut}(X,\Delta)$ a possibly trivial torus, $v_0 \in \mathrm{Val}_X^{T,\circ}$ a quasi-monomial valuation, and $g:\mathbb{R}\to\mathbb{R}_{>0}$ a continuous function. Assume that $\delta_{g,T}(X,\Delta,v_0)\le 1$. Then $\mathrm{gr}_v R$ is a finitely generated $\mathbb{C}$-algebra for any $v\in\mathrm{Val}_X^{T,\circ}$ computing $\delta_{g,T}(X,\Delta,v_0)$.*

Sketch of the proof. Let $v\in\mathrm{Val}_X^{T,\circ}$ be a valuation computing $\delta_{g,T}(X,\Delta,v_0)$. Theorem 1.1 follows from three steps.

(1) By Theorem 3.1, which follows from Lemma 3.2, Definition-Lemma 3.3, and [Xu20, Theorem 1.1], v is quasi-monomial.
(2) If $\delta_{g,T}(X,\Delta,v_0)\le 1$, then v is a monomial lc place of a special $\mathbb{Q}$-complement $(X,\Delta+\Gamma)$ of (X,Δ) (Proposition 4.1), which follows from Lemmas 4.3, 4.4, 4.5, and 4.2 and [BLXZ23, Lemma 4.2].
(3) For an integer $q\ge 1$ with $qr\in\mathbb{Z}$, [LXZ22, Theorem 4.2] implies that the q-th Veronese subring of $\mathrm{gr}_v R$ is finitely generated, and so is $\mathrm{gr}_v R$ by Lemma 5.1.

In the key new step, after a positive constant is subtracted from g, the weighted expected vanishing order becomes a convex combination of a weighted and an unweighted one (Lemma 4.5), and an auxiliary $\mathbb{Q}$-divisor in the unweighted part moves in a linear system without changing the value at v. The proof does not show that a valuation computing $\delta_{g,T}(X,\Delta,v_0)$ exists and provides no bound on the generators of $\mathrm{gr}_v R$. We use [LXZ22, Theorem 4.2] directly, Theorem 3.1 adapts [BJ20, Proposition 4.8(ii)], and the new constructions are in the proof of Proposition 4.1. Only **Steps 3 and 4** of that proof use $\delta_{g,T}(X,\Delta,v_0)\le 1$, **Step 3** for the ampleness of $-(K_X+\Delta+B)$ in Lemma 4.2 and **Step 4** to keep the coefficients of the moving $\mathbb{Q}$-divisor on a log resolution less than 1, and only [BLXZ23, Lemma 4.2] uses that v_0 is quasi-monomial.

(a) Splitting off a constant from g. The weighted basis type divisors in [BLXZ23, Proof of Theorem 5.4] need one basis of $H^0(X,\mathcal{O}_X(mL))$ compatible with v_0, v, and an auxiliary $\mathbb{Q}$-divisor (see [BLXZ23, Definition 4.1 and Lemma 5.6]), which does not exist in general. Lemma 4.5 uses two bases instead. Since the minimum c_1 of g on the interval $[0,\lambda_{\max}(\mathcal{F}_{v_0})]$, which contains the numbers $v_0(s)/(mr)$, is positive, $h=g-c_1/2\ge c_1/2$ there, and

$$S_{g,m}(v_0;w) = (1-t_m)S_{h,m}(v_0;w) + t_m S_m(w) \quad \text{for any } w\in\mathrm{Val}_X^\circ,$$

where $c_1/(2c_2)\le t_m<1$ for the maximum c_2 of g on the interval (Lemma 4.5 and (4.12) in the proof of Lemma 4.5, with S_m and $S_{h,m}$ as in Definition-Propositions 2.12 and 2.15).

By Lemma 2.17 and [BJ20, Lemma 3.5], $D=(1-t_m)D_1+t_mD_2$ satisfies $w(D)\le S_{g,m}(v_0;w)$, with equality at $w=v$, for an h-weighted (m,v_0)-basis type divisor D_1 and an m-basis type divisor D_2 that are compatible with v but defined by different bases. The basis of D_1 is compatible with v_0 and v, and since the weights of D_2 do not depend on v_0, the basis of D_2 can be compatible with v and with a filtration defined by an auxiliary $\mathbb{Q}$-divisor.

(b) An auxiliary $\mathbb{Q}$-divisor moving in a linear system. Log canonicity can be read off from $\delta_{g,T}(X,\Delta,v_0)$, an infimum over T-invariant valuations, only for T-invariant boundaries (Lemma 4.4). However, specialness asks that the strict transform of Γ on a log smooth model (Y,E) be at least an ample $\mathbb{Q}$-divisor whose support contains no stratum of (Y,E) (Definition 2.8), and T-invariant divisors may contain strata: for $\mathbb{G}_m$ acting on $Y=\mathbb{P}^1$ by scaling and $E=\{0\}+\{\infty\}$, the only invariant prime divisors on Y are the strata 0 and ∞. In the proof of Proposition 4.1, the basis of D_2 is compatible with v and with divisibility by powers of a T-eigensection s_0 in a T-invariant linear system V (**Steps 1 and 2**). Log canonicity is proved at s_0, and after the coefficients are decreased slightly, Bertini's theorem passes the klt property to a general $s\in V$

(**Steps 3 and 4**). Replacing s_0 by s does not change the value at v, and the divisor of s provides the ample $\mathbb{Q}$-divisor asked for by specialness.

Since the constant η of Lemma 4.2 depends only on X, Δ, v, and a finite set containing the coefficients of the auxiliary $\mathbb{Q}$-divisor for all m and s, one degree m and one general $s \in V$ suffice (**Step 5**), although the open subset of V on which **Step 4** proves the klt property depends on m.

(c) Quasi-monomiality. The argument of [BJ20, Proposition 4.8(ii)] compares v with a valuation w computing $\mathrm{lct}(X, \Delta; \mathfrak{a}_\bullet(v))$ through the strict monotonicity of the expected vanishing order S (Definition-Proposition 2.12, [BJ20, Proposition 3.15]). Here the quasi-monomial valuation w given by [Xu20, Theorem 1.1] need not be T-invariant, so $\delta_{g,T}(X, \Delta, v_0)$ does not bound $A_{X,\Delta}(w)$ from below. Definition-Lemma 3.3 replaces w by the T-invariant valuation w^T, where $w^T(f) = \min_\chi w(f_\chi)$ for the decomposition $f = \sum_\chi f_\chi$ into T-eigenfunctions. Then w^T takes the values of w on T-invariant ideals, and w^T belongs to the closure of the T-orbit of w, so $A_{X,\Delta}(w^T) \leq A_{X,\Delta}(w)$ by lower semicontinuity. Moreover, w^T is Abhyankar, hence quasi-monomial. After the normalization $w(\mathfrak{a}_\bullet(v)) = 1$, we have $A_{X,\Delta}(w) \leq A_{X,\Delta}(v)$, and $v \leq w^T$ because $w^T(\mathfrak{a}_m(v)) = w(\mathfrak{a}_m(v)) \geq m$ for any m. Lemma 3.2 bounds $S_g(v_0; w^T) - S_g(v_0; v)$ from below by $(c_1/c_2)(S(w^T) - S(v))$, comparing the filtrations induced by v and w^T on each graded piece of the filtration of v_0 (Lemma 2.17), and $S(v) < S(w^T)$ if $v \neq w^T$ by [BJ20, Proposition 3.15]. Since v computes $\delta_{g,T}(X, \Delta, v_0)$ and $A_{X,\Delta}(w^T) \leq A_{X,\Delta}(v)$, we have $S_g(v_0; w^T) \leq S_g(v_0; v)$, and hence $w^T = v$.

Structure of the paper. In Section 2 we recall preliminary material and fix notation. In particular, we recall the weighted invariants of Blum, Liu, Xu, and Zhuang [BLXZ23, Section 4.1] and special complements [LXZ22, Definition 3.3], and we prove an upper bound for the values of valuations on weighted basis type divisors (Lemma 2.17). In Section 3 we prove that any valuation in $\mathrm{Val}_X^{T,\circ}$ computing $\delta_{g,T}(X, \Delta, v_0)$ is quasi-monomial (Theorem 3.1). In Section 4 we prove a uniform complement lemma, construct equivariant log smooth models, and then construct special complements (Proposition 4.1). In Section 5 we prove Theorem 1.1.

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2. Preliminaries

We adopt the standard notation and definitions for the MMP from [Sho92, KM98, BCHM10] and use them freely.

2.1. Pairs, valuations, and complements.

Definition 2.1. A *pair* (X, B) consists of a normal quasi-projective variety X and an $\mathbb{R}$-divisor $B \geq 0$ on X such that $K_X + B$ is $\mathbb{R}$-Cartier. If we do not require that $B \geq 0$, then we say that (X, B) is a *sub-pair*. Let $h : X' \to X$ be any log resolution of (X, B) and E a prime divisor on X', and write $K_{X'} + B' := h^*(K_X + B)$. Then we define

$$a(E, X, B) := 1 - \mathrm{mult}_E B'$$

as the *log discrepancy* of E with respect to (X, B). For any divisorial valuation $v = c\,\mathrm{ord}_E$ for some real number $c > 0$, we denote by $a(v, X, B) := ca(E, X, B)$.

For any (sub-)pair (X, B), we say that (X, B) is *(sub-)lc* if $a(E, X, B) \geq 0$ for any prime divisor E over X, and *(sub-)klt* if $a(E, X, B) > 0$ for any prime divisor E over X. An *lc place* of (X, B) is a prime divisor E over X such that $a(E, X, B) = 0$. An *lc center* of (X, B) is the center of an lc place of (X, B) on X.

Definition 2.2. A pair (X, B) is a *log Fano pair* if X is projective, B is a $\mathbb{Q}$-divisor, (X, B) is klt, and $-(K_X + B)$ is ample. A normal projective variety X is *of Fano type* if there exists a $\mathbb{Q}$-divisor $B \geq 0$ on X such that (X, B) is a log Fano pair.

Definition 2.3. Let X be a normal projective variety. We denote by Val_X the set of real valuations v of the field of rational functions on X such that $v(c) = 0$ for any $c \in \mathbb{C}^\times$, equipped with the weakest topology such that the map $\mathrm{Val}_X \to \mathbb{R}$, $v \to v(f)$ is continuous for any non-zero rational function f on X. The valuation v with $v(f) = 0$ for any non-zero rational function f on X is the *trivial valuation*. The *center* $\mathrm{center}_X v$ of $v \in \mathrm{Val}_X$ is the unique scheme-theoretic point $\xi \in X$ such that $v \geq 0$ on $\mathcal{O}_{X,\xi}$ and $v > 0$ on the maximal ideal of $\mathcal{O}_{X,\xi}$. For $v \in \mathrm{Val}_X$, a line bundle $\mathcal{L}$ on X, and $s \in H^0(X, \mathcal{L})\backslash\{0\}$, we set $v(s) := v(f)$, where f is the local function of s under a trivialization of $\mathcal{L}$ near $\mathrm{center}_X v$. For a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $D \geq 0$ on X, we set $v(D) := \frac{1}{m} v(f)$, where m is a positive integer such that mD is Cartier and f is a local equation of mD near $\mathrm{center}_X v$, and we extend $v(D)$ linearly to finite $\mathbb{R}$-linear combinations of $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisors. For a non-zero ideal sheaf $\mathfrak{a} \subset \mathcal{O}_X$, we set

$$v(\mathfrak{a}) := \min\{v(f) \mid f \in \mathfrak{a} \cdot \mathcal{O}_{X,\mathrm{center}_X v}\}.$$

For $u, w \in \mathrm{Val}_X$, we write $u \leq w$ if $\mathrm{center}_X w \in \overline{\{\mathrm{center}_X u\}}$ and $u(f) \leq w(f)$ for any $f \in \mathcal{O}_{X,\mathrm{center}_X w}$.

Definition 2.4. Let Y be a smooth variety and E a reduced divisor on Y with simple normal crossing support. A *stratum* of (Y, E) is an irreducible component of $E_1 \cap \cdots \cap E_k$ for some irreducible components $E_1, \ldots, E_k$ of E with $k \geq 1$. Let η be the generic point of a stratum of (Y, E), let $y_1, \ldots, y_k \in \mathcal{O}_{Y,\eta}$ be local equations of the irreducible components of E containing η, and fix an isomorphism

$$\widehat{\mathcal{O}}_{Y,\eta} \cong \kappa(\eta)[[y_1, \ldots, y_k]].$$

For $\alpha \in \mathbb{R}^k_{\geq 0}$, the valuation v_α is defined by

$$v_\alpha(f) := \min\{\langle \alpha, \beta \rangle \mid c_\beta \neq 0\} \quad \text{for any non-zero } f = \sum_{\beta \in \mathbb{N}^k} c_\beta y^\beta \in \mathcal{O}_{Y,\eta},$$

where $c_\beta \in \kappa(\eta)$. We denote by $\mathrm{QM}_\eta(Y, E)$ the set of the valuations v_α, $\alpha \in \mathbb{R}^k_{\geq 0}$, and by $\mathrm{QM}(Y, E)$ the union of $\mathrm{QM}_\eta(Y, E)$ over the generic points η of the strata of (Y, E) [JM12, Section 3.1].

Let (X, B) be a pair such that X is projective. A *log smooth model* $\pi : (Y, E) \to (X, B)$ consists of a log resolution $\pi : Y \to X$ of (X, B) and a reduced divisor E on Y such that $E + \mathrm{Exc}(\pi) + \pi_*^{-1} B$ has simple normal crossing support. A valuation $v \in \mathrm{Val}_X$ is *quasi-monomial* if $v \in \mathrm{QM}(Y, E)$ for some log smooth model (Y, E) of (X, B), and v is *Abhyankar* if $\mathrm{rat.rk}(v) + \mathrm{tr.deg}(v) = \dim X$, where $\mathrm{rat.rk}(v)$ is the rank of the value group of v and $\mathrm{tr.deg}(v)$ is the transcendence degree of the residue field of the valuation ring of v [LXZ22, Definitions 2.1 and 2.4] (see also [ELS03, Section 2]).

Remark 2.5. Whether v is quasi-monomial does not depend on B. Let $\pi : (Y, E) \to (X, B)$ be a log smooth model, $v \in \mathrm{QM}(Y, E)$, and B' another boundary on X. Let F be the reduced divisor on Y with support $\mathrm{Supp}\, E \cup \mathrm{Exc}(\pi)$. By the simple normal crossing condition, no irreducible component of F that is not a component of E contains a stratum of (Y, E), so $\mathrm{QM}(Y, E) \subset \mathrm{QM}(Y, F)$. Let $h : Y' \to Y$ be a log resolution of $(Y, F + \pi_*^{-1} B')$ and E' the reduced divisor on Y' with support $h^{-1}(\mathrm{Supp}\, F) \cup \mathrm{Exc}(h)$. Then (Y', E') is a log smooth model of (X, B') dominating (Y, F), and $\mathrm{QM}(Y, F) \subset \mathrm{QM}(Y', E')$ [JM12, Lemma 4.6 and Corollary 5.4] (applied to the log smooth pairs (Y, F) and (Y', E') over Y in the sense of [JM12, Definition 3.2]). By [JM12, Remark 3.4] and **Step 5** of the proof of Definition-Lemma 3.3, a valuation $v \in \mathrm{Val}_X$ is quasi-monomial in the sense of Definition 2.4 if and only if v is quasi-monomial in the sense of [LXZ22, Definition 2.4].

Definition 2.6. Let (X, B) be a pair such that X is projective. For $v \in \mathrm{Val}_X$, we denote by $A_{X,B}(v) \in \mathbb{R} \cup \{+\infty\}$ the log discrepancy of v with respect to (X, B) [LXZ22, Definition 2.5] (see [JM12, Section 5] for X smooth and $B = 0$), so that $A_{X,B}(v) = a(v, X, B)$ if v is divisorial. We set

$$\mathrm{Val}_X^\circ := \{v \in \mathrm{Val}_X \mid v \text{ is non-trivial and } A_{X,B}(v) < +\infty\}.$$

If a torus T acts on (X, B), then T acts on the rational functions on X by $t \cdot f := f \circ t^{-1}$ and on Val_X by $(t \cdot v)(f) := v(f \circ t)$. A non-zero rational function f on X is a *T-eigenfunction of weight* $\chi \in \mathrm{Hom}(T, \mathbb{G}_m)$ if $t \cdot f = \chi(t) f$ for any $t \in T$, and $v \in \mathrm{Val}_X$ is *T-invariant* if $t \cdot v = v$ for any $t \in T$. We denote by Val_X^T the set of T-invariant valuations in Val_X, and by $\mathrm{Val}_X^{T,\circ}$ the set of T-invariant valuations in Val_X° [BLXZ23, Definition 2.1].

In [LXZ22, Definition 2.5], B is a $\mathbb{Q}$-divisor, and the formulas there define $A_{X,B}$ also for $\mathbb{R}$-divisors B. The pair (X, B) is omitted from the notation Val_X°. From Set-up 2.10 on, Val_X° and $\mathrm{Val}_X^{T,\circ}$ are defined with respect to (X, Δ).

Definition 2.7. Let (X, B) be a pair such that X is projective, and let $v \in \mathrm{Val}_X$. For $\lambda \in \mathbb{R}_{\geq 0}$, the *valuation ideal* $\mathfrak{a}_\lambda(v) \subset \mathcal{O}_X$ is the ideal sheaf whose stalk at a point $x \in X$ is $\{f \in \mathcal{O}_{X,x} \mid v(f) \geq \lambda\}$ if $x \in \overline{\{\mathrm{center}_X v\}}$, and $\mathcal{O}_{X,x}$ otherwise [BJ20, Section 1.2]. We write $\mathfrak{a}_\bullet(v)$ for the graded sequence $(\mathfrak{a}_m(v))_{m \geq 1}$. For a graded sequence $\mathfrak{a}_\bullet = (\mathfrak{a}_m)_{m \geq 1}$ of non-zero ideal sheaves on X and $w \in \mathrm{Val}_X$, we set

$$w(\mathfrak{a}_\bullet) := \inf_{m \geq 1} \frac{w(\mathfrak{a}_m)}{m}.$$

The *log canonical threshold* of $\mathfrak{a}_\bullet$ with respect to (X, B) is

$$\mathrm{lct}(X, B; \mathfrak{a}_\bullet) := \inf \left\{ \frac{A_{X,B}(w)}{w(\mathfrak{a}_\bullet)} \;\middle|\; w \in \mathrm{Val}_X^\circ,\ w(\mathfrak{a}_\bullet) > 0 \right\}$$

[Xu20, Section 2.2.1]. A valuation $w \in \mathrm{Val}_X^\circ$ with $w(\mathfrak{a}_\bullet) > 0$ *computes* $\mathrm{lct}(X, B; \mathfrak{a}_\bullet)$ if the infimum is attained at w.

In [Xu20, Section 2.2.1], $\mathrm{lct}(X, B; \mathfrak{a}_\bullet)$ is defined for klt pairs by means of the ideals $\mathfrak{a}_m$, and the expression of $\mathrm{lct}(X, B; \mathfrak{a}_\bullet)$ as an infimum over Val_X° in Definition 2.7 is quoted there from [JM12, Corollary 6.9].

Definition 2.8. Let (X, B) be a pair such that X is projective and B is a $\mathbb{Q}$-divisor. For a $\mathbb{Q}$-divisor $\Gamma \geq 0$ on X, we say that $(X, B + \Gamma)$ is a *$\mathbb{Q}$-complement* of (X, B) if $(X, B + \Gamma)$ is lc and $K_X + B + \Gamma \sim_{\mathbb{Q}} 0$. Assume that (X, B) is a log Fano pair, and let $\pi : (Y, E) \to (X, B)$ be a log smooth model and $(X, B + \Gamma)$ a $\mathbb{Q}$-complement of (X, B). We say that $(X, B + \Gamma)$ is *special* with respect to (Y, E) if $\pi_*^{-1}\Gamma \geq G$ for some ample $\mathbb{Q}$-divisor $G \geq 0$ on Y whose support does not contain any stratum of (Y, E). A non-trivial valuation $v \in \mathrm{QM}(Y, E)$ is a *monomial lc place* of $(X, B + \Gamma)$ with respect to (Y, E) if $A_{X,B+\Gamma}(v) = 0$ [LXZ22, Definitions 2.27 and 3.3].

In [LXZ22, Definition 3.3], monomial lc places are normalized by $A_{X,B}(v) = 1$. In Definition 2.8, we do not require $A_{X,B}(v) = 1$.

Remark 2.9. For a log Fano pair (X, B), a $\mathbb{Q}$-complement $(X, B + \Gamma)$ of (X, B), and a quasi-monomial valuation v with $A_{X,B+\Gamma}(v) = 0$, the associated graded ring of v need not be finitely generated. Indeed, if $C \subset \mathbb{P}^2$ is an irreducible nodal cubic curve and v_t is the quasi-monomial valuation with weights $(1, t)$ along the analytic branches C_1 and C_2 of C at the node for a real number $t > 0$, then $(\mathbb{P}^2, C)$ is a $\mathbb{Q}$-complement of $(\mathbb{P}^2, 0)$, $A_{\mathbb{P}^2,C}(v_t) = 0$, and the associated graded ring of the section ring $\bigoplus_{m \geq 0} H^0(\mathbb{P}^2, -mK_{\mathbb{P}^2})$ with respect to v_t (defined as in Definition 2.18) is finitely generated if and only if $t \in \mathbb{Q}$ or $(7 - 3\sqrt{5})/2 < t < (7 + 3\sqrt{5})/2$ [LXZ22, Theorem 6.1]. The condition that $\pi_*^{-1}\Gamma \geq G$ for some ample $\mathbb{Q}$-divisor $G \geq 0$ on Y whose support does not contain any stratum of (Y, E) is the hypothesis under which [LXZ22, Theorem 4.2] implies finite generation, and Section 5 uses it in this way.

2.2. **Filtrations and weighted stability thresholds.** In this subsection we recall filtrations of section rings, basis type divisors, and the weighted stability thresholds of [BLXZ23].

Set-up 2.10. Let (X,Δ), d, r, L, R_m, N_m, R, T, M, v_0, and g be as follows.

(1) (X,Δ) is a log Fano pair of dimension d.
(2) r is a positive real number such that $L := -r(K_X+\Delta)$ is Cartier.
(3) $R_m := H^0(X,\mathcal{O}_X(mL))$ and $N_m := \dim R_m$ for $m\in\mathbb{N}$, and $R := \bigoplus_{m\in\mathbb{N}} R_m$.
(4) $T\subset \mathrm{Aut}(X,\Delta)$ is a possibly trivial torus, and $M := \mathrm{Hom}(T,\mathbb{G}_m)$ is the character lattice of T.
(5) $v_0\in \mathrm{Val}_X^{T,\circ}$ is a quasi-monomial valuation.
(6) $g:\mathbb{R}\to\mathbb{R}_{>0}$ is a continuous function.

Definition 2.11. Notation as in Set-up 2.10. A *filtration* $\mathcal{F}$ of R is a collection of linear subspaces $\mathcal{F}^\lambda R_m\subset R_m$, $\lambda\in\mathbb{R}$, $m\in\mathbb{N}$, such that

$$\mathcal{F}^\lambda R_m = \bigcap_{\lambda'<\lambda} \mathcal{F}^{\lambda'} R_m \quad\text{and}\quad \mathcal{F}^\lambda R_m\cdot \mathcal{F}^{\lambda'} R_{m'} \subset \mathcal{F}^{\lambda+\lambda'} R_{m+m'}$$

for any $\lambda,\lambda'\in\mathbb{R}$ and $m,m'\in\mathbb{N}$, $\mathcal{F}^\lambda R_m\subset \mathcal{F}^{\lambda'} R_m$ if $\lambda\geq\lambda'$, $\mathcal{F}^\lambda R_m = R_m$ for $\lambda\ll 0$, and $\mathcal{F}^\lambda R_m = 0$ for $\lambda\gg 0$ [BLXZ23, Definition 2.2]. We set

$$\mathcal{F}^{>\lambda} R_m := \bigcup_{\lambda'>\lambda} \mathcal{F}^{\lambda'} R_m \quad\text{and}\quad \mathrm{gr}^\lambda_{\mathcal{F}} R_m := \mathcal{F}^\lambda R_m/\mathcal{F}^{>\lambda} R_m.$$

For another filtration $\mathcal{G}$ of R and $t\in\mathbb{R}$, we denote by $\mathcal{G}^t\,\mathrm{gr}^\lambda_{\mathcal{F}} R_m$ the image of $\mathcal{F}^\lambda R_m\cap \mathcal{G}^t R_m$ in $\mathrm{gr}^\lambda_{\mathcal{F}} R_m$. A basis $(s_1,\dots,s_{N_m})$ of R_m is *compatible* with $\mathcal{F}$ if $\mathcal{F}^\lambda R_m$ is spanned by $\{s_i\mid s_i\in \mathcal{F}^\lambda R_m\}$ for any $\lambda\in\mathbb{R}$. The filtration $\mathcal{F}$ is *linearly bounded* if there exists a real number $c>0$ such that $\mathcal{F}^{mc} R_m = 0$ for any positive integer m, and in this case we set

$$\lambda_{\max}(\mathcal{F}) := \sup\left\{ \frac{\max\{\lambda\in\mathbb{R}\mid \mathcal{F}^\lambda R_m\neq 0\}}{mr} \;\middle|\; m\in\mathbb{N}^+,\ R_m\neq 0\right\}.$$

Definition-Proposition 2.12. Notation as in Set-up 2.10. For $v\in\mathrm{Val}_X$, the filtration $\mathcal{F}_v$ of R is defined by

$$\mathcal{F}_v^\lambda R_m := \{s\in R_m\backslash\{0\}\mid v(s)\geq\lambda\}\cup\{0\}$$

[BLXZ23, Example 2.1]. For a positive integer m with $R_m\neq 0$, an *m-basis type divisor* is a divisor of the form $\frac{1}{mrN_m}\sum_{i=1}^{N_m}\mathrm{div}(s_i)$ for a basis $(s_1,\dots,s_{N_m})$ of R_m [FO18, Definition 0.1], and it is *compatible* with a filtration $\mathcal{F}$ of R if the basis is compatible with $\mathcal{F}$. For a linearly bounded filtration $\mathcal{F}$ of R, we set

$$S_m(\mathcal{F}) := \frac{1}{mrN_m}\sum_{i=1}^{N_m}\max\{\lambda\in\mathbb{R}\mid s_i\in\mathcal{F}^\lambda R_m\}$$

for a basis $(s_1,\dots,s_{N_m})$ of R_m compatible with $\mathcal{F}$, and $S(\mathcal{F}) := \lim_{m\to+\infty} S_m(\mathcal{F})$ [LXZ22, Definition 2.20] (after a translation, see [BLXZ23, Section 2.1], we may assume $\mathcal{F}^0 R_m = R_m$, as in [LXZ22, Definition 2.20]). For $v\in\mathrm{Val}_X^\circ$, the filtration $\mathcal{F}_v$ is linearly bounded [BLXZ23, Example 2.1(1)], and we write $S_m(v) := S_m(\mathcal{F}_v)$ and $S(v) := S(\mathcal{F}_v)$.

In [BJ20, Section 2.6], the jumping numbers of filtrations of R_m are divided by m instead of mr. Thus, for the line bundle $\mathcal{O}_X(L)$, the invariants S_m, S, and T and the m-basis type divisors of [BJ20] are r times our S_m, S, $\lambda_{\max}$, and m-basis type divisors.

Definition-Proposition 2.13. Notation as in Set-up 2.10. Then $r\in\mathbb{Q}$, and we denote by q the smallest positive integer such that $qr\in\mathbb{Z}$. For any positive integer m divisible by q, the action of T on (X,Δ) induces a linear action $(t,s)\to t\cdot s$ of T on $R_m = H^0(X,\mathcal{O}_X(-mr(K_X+\Delta)))$, and the following hold.

(1) $t\cdot(s_1s_2)=(t\cdot s_1)(t\cdot s_2)$, $t\cdot(fs_1)=(t\cdot f)(t\cdot s_1)$, and $\mathrm{div}(t\cdot s_1)=t(\mathrm{div}(s_1))$ for any $t\in T$, $s_1\in R_m$, $s_2\in R_{m'}$ with $q\mid m'$, and any rational function f on X such that $fs_1\in R_m$, where $t\cdot f$ is as in Definition 2.6.
(2) $R_m=\bigoplus_{\chi\in M}R_{m,\chi}$, where $R_{m,\chi}$ is the set of $s\in R_m$ with $t\cdot s=\chi(t)s$ for any $t\in T$, and $W=\bigoplus_{\chi\in M}(W\cap R_{m,\chi})$ for any T-invariant linear subspace $W\subset R_m$. We call the non-zero elements of $R_{m,\chi}$ the *T-eigensections* of weight χ.
(3) $\mathcal{F}_v^\lambda R_m$ is T-invariant for any $v\in\mathrm{Val}_X^T$ and $\lambda\in\mathbb{R}$.
(4) For any two filtrations $\mathcal{F}$ and $\mathcal{G}$ of R such that $\mathcal{F}^\lambda R_m$ and $\mathcal{G}^\lambda R_m$ are T-invariant for any $\lambda\in\mathbb{R}$, there exists a basis of R_m that consists of T-eigensections and is compatible with $\mathcal{F}$ and $\mathcal{G}$.

Proof. Since v_0 is non-trivial, $d\geq 1$, and hence there exists a curve C on X. Since L is Cartier and $-(K_X+\Delta)$ is an ample $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor,

$$r=\frac{(L\cdot C)}{(-(K_X+\Delta)\cdot C)}\in\mathbb{Q}.$$

Let m be a positive integer divisible by q. Then $mr\in\mathbb{N}^+$, $mL=-mr(K_X+\Delta)$, and R_m is identified with the space of rational sections σ of $\omega_X^{[-mr]}$ such that $\mathrm{div}(\sigma)\geq mr\Delta$, so that $\mathrm{div}(s)=\mathrm{div}(\sigma)-mr\Delta$ for the element $s\in R_m$ corresponding to σ. The group T acts on the rational sections of the sheaves $\omega_X^{[-k]}$, $k\in\mathbb{Z}$, by $t\cdot\sigma:=(t^{-1})^*\sigma$. For $k=0$, this is the action $t\cdot f=f\circ t^{-1}$ of Definition 2.6 on rational functions. The action is compatible with the multiplication maps

$$\omega_X^{[-k]}\otimes\omega_X^{[-k']}\to\omega_X^{[-k-k']},$$

and

$$\mathrm{div}(t\cdot\sigma)=t(\mathrm{div}(\sigma)).$$

Since $t(\Delta)=\Delta$ for any $t\in T$, the action preserves R_m, and

$$\mathrm{div}(t\cdot s)=t(\mathrm{div}(\sigma))-mr\Delta=t(\mathrm{div}(s)).$$

Thus (1) holds. Since the action of T on X is algebraic, R_m is a rational representation of T (the action of T induces a T-linearization of the invertible sheaf $\mathcal{O}_X(mL)$, and the global sections of a T-linearized invertible sheaf form a rational T-module, see, e.g., [MFK94, Chapter 1, Section 3]). Since T is diagonalizable, R_m and any T-invariant linear subspace $W\subset R_m$, which is itself a rational representation of T, are direct sums of weight spaces [Bor91, Proposition 8.4]. This implies (2).

For $v\in\mathrm{Val}_X^T$ and $s\in R_m\backslash\{0\}$, (1) and the T-invariance of v imply

$$v(t\cdot s)=v(t(\mathrm{div}(s)))=v(\mathrm{div}(s))=v(s),$$

which implies (3). For $\mathcal{F}$ and $\mathcal{G}$ as in (4), (2) implies

$$\mathcal{F}^\lambda R_m=\bigoplus_{\chi\in M}(\mathcal{F}^\lambda R_m\cap R_{m,\chi})\quad\text{and}\quad\mathcal{G}^\lambda R_m=\bigoplus_{\chi\in M}(\mathcal{G}^\lambda R_m\cap R_{m,\chi})$$

for any $\lambda\in\mathbb{R}$. For any $\chi\in M$, the subspaces $\mathcal{F}^\lambda R_m\cap R_{m,\chi}$ and $\mathcal{G}^\lambda R_m\cap R_{m,\chi}$, $\lambda\in\mathbb{R}$, are two filtrations of $R_{m,\chi}$, and hence there exists a basis of $R_{m,\chi}$ compatible with both [AZ22, Lemma 3.1]. The union over $\chi\in M$ of the bases of $R_{m,\chi}$ is a basis of R_m as in (4). □

Definition-Proposition 2.14. Notation as in Set-up 2.10. Let $\mathcal{F}_0$ and $\mathcal{F}_1$ be linearly bounded filtrations of R. For a positive integer m with $R_m\neq 0$, let $(s_1,\dots,s_{N_m})$ be a basis of R_m compatible with $\mathcal{F}_0$ and $\mathcal{F}_1$, and set

$$\boldsymbol{\nu}_m:=\frac{1}{N_m}\sum_{i=1}^{N_m}\boldsymbol{\delta}_{(\lambda_{0,i}/(mr),\lambda_{1,i}/(mr))},$$

where $\lambda_{j,i}$ is the largest real number λ such that $s_i \in \mathcal{F}_j^\lambda R_m$, and $\boldsymbol{\delta}_p$ is the Dirac measure at $p \in \mathbb{R}^2$. The *compatible Duistermaat–Heckman measure* $\mathrm{DH}_{\mathcal{F}_0,\mathcal{F}_1}$ is the weak limit of $\boldsymbol{\nu}_m$ as $m \to +\infty$, which is a compactly supported probability measure on $\mathbb{R}^2$ [BLXZ23, Section 3.1.3 and Theorem 3.3].

Definition-Proposition 2.15. Notation as in Set-up 2.10. Let $\varphi : \mathbb{R} \to \mathbb{R}_{>0}$ be a continuous function and m a positive integer such that $R_m \neq 0$. For a basis $(s_1, \dots, s_{N_m})$ of R_m compatible with $\mathcal{F}_{v_0}$, we set

$$Q_{\varphi,m} := \sum_{i=1}^{N_m} \varphi\Big(\frac{v_0(s_i)}{mr}\Big),$$

which does not depend on the basis (the multiset $\{v_0(s_i)\}$ is determined by the dimensions of the spaces $\mathcal{F}_{v_0}^\lambda R_m$, see [BLXZ23, Section 2.1.1]), and we call

$$D := \frac{1}{mrQ_{\varphi,m}} \sum_{i=1}^{N_m} \varphi\Big(\frac{v_0(s_i)}{mr}\Big) \operatorname{div}(s_i)$$

a *φ-weighted (m, v_0)-basis type divisor* [BLXZ23, Definition 4.1]. We say that D is *compatible* with a filtration $\mathcal{F}$ if $(s_1, \dots, s_{N_m})$ is compatible with $\mathcal{F}$. For a linearly bounded filtration $\mathcal{F}$ of R, let $\boldsymbol{\nu}_m$ be the measures of Definition-Proposition 2.14 for $\mathcal{F}_0 = \mathcal{F}_{v_0}$ and $\mathcal{F}_1 = \mathcal{F}$, let $\boldsymbol{\nu}$ be their weak limit, and set

$$S_{\varphi,m}(v_0;\mathcal{F}) := \frac{\int_{\mathbb{R}^2} \varphi(x) y d\boldsymbol{\nu}_m}{\int_{\mathbb{R}^2} \varphi(x) d\boldsymbol{\nu}_m} \quad \text{and} \quad S_\varphi(v_0;\mathcal{F}) := \lim_{m\to+\infty} S_{\varphi,m}(v_0;\mathcal{F}) = \frac{\int_{\mathbb{R}^2} \varphi(x) y d\boldsymbol{\nu}}{\int_{\mathbb{R}^2} \varphi(x) d\boldsymbol{\nu}}.$$

For $w \in \mathrm{Val}_X^\circ$, we write $S_{\varphi,m}(v_0;w) := S_{\varphi,m}(v_0;\mathcal{F}_w)$ and $S_\varphi(v_0;w) := S_\varphi(v_0;\mathcal{F}_w)$. The *weighted stability threshold* is

$$\delta_{g,T}(X,\Delta,v_0) := \inf_{v \in \mathrm{Val}_X^{T,\circ}} \frac{A_{X,\Delta}(v)}{S_g(v_0;v)},$$

and $v \in \mathrm{Val}_X^{T,\circ}$ *computes* $\delta_{g,T}(X,\Delta,v_0)$ if

$$\delta_{g,T}(X,\Delta,v_0) = \frac{A_{X,\Delta}(v)}{S_g(v_0;v)}$$

[BLXZ23, Section 4.1]. We denote by c_1 and c_2 the minimum and the maximum of g on the interval $[0, \lambda_{\max}(\mathcal{F}_{v_0})]$.

Since $0 \le v_0(s)/(mr) \le \lambda_{\max}(\mathcal{F}_{v_0})$ for any positive integer m with $R_m \neq 0$ and any $s \in R_m \backslash \{0\}$, $S_{\varphi,m}(v_0;\mathcal{F})$ only depends on the restriction of φ to $[0, \lambda_{\max}(\mathcal{F}_{v_0})]$, which is why c_1 and c_2 only involve the restriction of g to $[0, \lambda_{\max}(\mathcal{F}_{v_0})]$.

Lemma 2.16. *Notation as in Set-up 2.10. Let $\mathcal{F}$ be a linearly bounded filtration of R such that $\mathcal{F}^0 R_m = R_m$ for any $m \in \mathbb{N}$. Then*

$$S_{g,m}(v_0;\mathcal{F}) \ge \frac{c_1}{c_2} S_m(\mathcal{F}) \quad \text{and} \quad S_g(v_0;\mathcal{F}) \ge \frac{c_1}{c_2} S(\mathcal{F})$$

for any positive integer m with $R_m \neq 0$. Moreover, $0 < S_g(v_0;w) < +\infty$ for any $w \in \mathrm{Val}_X^\circ$.

Proof. Let m be a positive integer with $R_m \neq 0$, and let $(s_1, \dots, s_{N_m})$ be a basis of R_m compatible with $\mathcal{F}_{v_0}$ and $\mathcal{F}$ [AZ22, Lemma 3.1]. Let $\lambda_{0,i}$ and $\lambda_{1,i}$ be as in Definition-Proposition 2.14 for $\mathcal{F}_0 = \mathcal{F}_{v_0}$ and $\mathcal{F}_1 = \mathcal{F}$. Then $\lambda_{0,i} = v_0(s_i) \ge 0$ since v_0 has a center on X, $\lambda_{0,i} \le mr\lambda_{\max}(\mathcal{F}_{v_0})$ by Definition 2.11, and $\lambda_{1,i} \ge 0$ since $\mathcal{F}^0 R_m = R_m$. Thus $c_1 \le g(\lambda_{0,i}/(mr)) \le c_2$ for any i, and

$$S_{g,m}(v_0;\mathcal{F}) = \frac{\sum_{i=1}^{N_m} g(\lambda_{0,i}/(mr))\lambda_{1,i}}{mr\sum_{i=1}^{N_m} g(\lambda_{0,i}/(mr))} \ge \frac{c_1 \sum_{i=1}^{N_m} \lambda_{1,i}}{c_2 mr N_m} = \frac{c_1}{c_2} S_m(\mathcal{F}).$$

Letting $m \to +\infty$, Definition-Propositions 2.12 and 2.15 imply $S_g(v_0;\mathcal{F}) \ge \frac{c_1}{c_2} S(\mathcal{F})$.

Let $w \in \mathrm{Val}_X^\circ$. Since w has a center on X, $w(s) \geq 0$ for any $s \in R_m \backslash \{0\}$, that is, $\mathcal{F}_w^0 R_m = R_m$ for any $m \in \mathbb{N}$. Moreover, $\mathcal{F}_w$ is linearly bounded (Definition-Proposition 2.12). Thus

$$S_g(v_0; w) \geq \frac{c_1}{c_2} S(w) \geq \frac{c_1}{c_2} \cdot \frac{\lambda_{\max}(\mathcal{F}_w)}{d+1},$$

where the second inequality is [BJ20, Lemmas 2.6 and 2.9]. Let $\xi := \mathrm{center}_X w$. If ξ is the generic point of X, then $w \geq 0$ on the field of rational functions on X, and $w(f) + w(f^{-1}) = 0$ implies that w is trivial. Thus the closure of ξ is a proper closed subset of X with non-zero ideal sheaf $\mathcal{I} \neq \mathcal{O}_X$. Since L is ample, $\mathcal{I} \otimes \mathcal{O}_X(mL)$ is globally generated for some positive integer m, and hence $H^0(X, \mathcal{I} \otimes \mathcal{O}_X(mL)) \subset R_m$ contains a non-zero element s. The local function of s at ξ belongs to the maximal ideal of $\mathcal{O}_{X,\xi}$, so $w(s) > 0$ by Definition 2.3. Hence $\lambda_{\max}(\mathcal{F}_w) \geq w(s)/(mr) > 0$. Since $c_1 > 0$, we obtain $S_g(v_0; w) > 0$.

Finally, for any positive integer m with $R_m \neq 0$ and a basis of R_m compatible with $\mathcal{F}_{v_0}$ and $\mathcal{F}_w$, we have $\lambda_{1,i}/(mr) \leq \lambda_{\max}(\mathcal{F}_w)$ for any i, where $\lambda_{1,i}$ is as in Definition-Proposition 2.14 for $\mathcal{F}_1 = \mathcal{F}_w$. Since $g > 0$, $S_{g,m}(v_0; w) \leq \lambda_{\max}(\mathcal{F}_w)$. Letting $m \to +\infty$, we obtain $S_g(v_0; w) \leq \lambda_{\max}(\mathcal{F}_w)$. If $c > 0$ is a real number such that $\mathcal{F}_w^{mc} R_m = 0$ for any positive integer m, then $\lambda_{\max}(\mathcal{F}_w) \leq c/r$. Thus $S_g(v_0; w) < +\infty$. □

Lemmas 3.2 and 4.5 are based on Lemma 2.17, which only involves the filtrations induced by $\mathcal{F}_w$ on the graded pieces $\mathrm{gr}^\lambda_{\mathcal{F}_{v_0}} R_m$.

Lemma 2.17. *Notation as in Set-up 2.10. Let $\varphi : \mathbb{R} \to \mathbb{R}_{>0}$ be a continuous function, m a positive integer such that $R_m \neq 0$, and $w \in \mathrm{Val}_X^\circ$. Then*

$$S_{\varphi,m}(v_0; w) = \frac{1}{Q_{\varphi,m}} \sum_{\lambda \in \mathbb{R}} \varphi\Big(\frac{\lambda}{mr}\Big) \int_0^{+\infty} \dim \mathcal{F}_w^{mrt} \mathrm{gr}^\lambda_{\mathcal{F}_{v_0}} R_m dt.$$

Moreover, $w(D) \leq S_{\varphi,m}(v_0; w)$ for any φ-weighted (m, v_0)-basis type divisor D, and equality holds if D is compatible with $\mathcal{F}_w$.

Proof. For $\lambda \in \mathbb{R}$, we write $V_\lambda := \mathrm{gr}^\lambda_{\mathcal{F}_{v_0}} R_m$ and denote by $\bar{s}$ the image in V_λ of $s \in \mathcal{F}^\lambda_{v_0} R_m$. Only finitely many V_λ are non-zero. Let $(s_1, \ldots, s_{N_m})$ be a basis of R_m compatible with $\mathcal{F}_{v_0}$, D the corresponding φ-weighted (m, v_0)-basis type divisor, and $I_\lambda := \{i \mid v_0(s_i) = \lambda\}$. Since $\mathcal{F}^\lambda_{v_0} R_m$ and $\mathcal{F}^{>\lambda}_{v_0} R_m$ are spanned by $\{s_i \mid v_0(s_i) \geq \lambda\}$ and $\{s_i \mid v_0(s_i) > \lambda\}$ respectively, $\{\bar{s}_i \mid i \in I_\lambda\}$ is a basis of V_λ. Since $w(\mathrm{div}(s_i)) = w(s_i)$,

$$w(D) = \frac{1}{Q_{\varphi,m}} \sum_{\lambda \in \mathbb{R}} \varphi\Big(\frac{\lambda}{mr}\Big) \sum_{i \in I_\lambda} \frac{w(s_i)}{mr}.$$

Assume that $(s_1, \ldots, s_{N_m})$ is also compatible with $\mathcal{F}_w$, which is possible by [AZ22, Lemma 3.1]. Then $\mathcal{F}^\lambda_{v_0} R_m \cap \mathcal{F}_w^{mrt} R_m$ is spanned by $\{s_i \mid v_0(s_i) \geq \lambda,\ w(s_i) \geq mrt\}$, and hence $\mathcal{F}_w^{mrt} V_\lambda$ is spanned by the linearly independent elements $\bar{s}_i$, $i \in I_\lambda$, $w(s_i) \geq mrt$. Since $w(s_i) \geq 0$,

$$\int_0^{+\infty} \dim \mathcal{F}_w^{mrt} V_\lambda dt = \int_0^{+\infty} \#\{i \in I_\lambda \mid w(s_i) \geq mrt\} dt = \sum_{i \in I_\lambda} \frac{w(s_i)}{mr}.$$

Moreover, $\lambda_{0,i} = v_0(s_i)$ and $\lambda_{1,i} = w(s_i)$ in Definition-Proposition 2.14 for $\mathcal{F}_0 = \mathcal{F}_{v_0}$ and $\mathcal{F}_1 = \mathcal{F}_w$, so Definition-Proposition 2.15 implies

$$S_{\varphi,m}(v_0; w) = \frac{1}{Q_{\varphi,m}} \sum_{i=1}^{N_m} \varphi\Big(\frac{v_0(s_i)}{mr}\Big) \frac{w(s_i)}{mr} = w(D).$$

Combining the last three formulas, we obtain the formula of the lemma and the equality for D compatible with $\mathcal{F}_w$.

Now let $(s_1, \dots, s_{N_m})$ be any basis of R_m compatible with $\mathcal{F}_{v_0}$. For $\lambda \in \mathbb{R}$ and $e \in V_\lambda \backslash \{0\}$, let $\mu(e)$ be the largest real number t such that $e \in \mathcal{F}_w^{mrt} V_\lambda$. The number $\mu(e)$ exists because $\mathcal{F}_w^{mrt} R_m$ takes finitely many values and $\mathcal{F}_w^{mrt} R_m = \bigcap_{t'<t} \mathcal{F}_w^{mrt'} R_m$. For $i \in I_\lambda$, we have

$$s_i \in \mathcal{F}_{v_0}^{\lambda} R_m \cap \mathcal{F}_w^{w(s_i)} R_m,$$

so $w(s_i)/(mr) \le \mu(\bar{s}_i)$. For any $t \ge 0$, the elements $\bar{s}_i$ with $i \in I_\lambda$ and $\mu(\bar{s}_i) \ge t$ are linearly independent in $\mathcal{F}_w^{mrt} V_\lambda$. Since $\mu(\bar{s}_i) \ge w(s_i)/(mr) \ge 0$, we obtain

$$\sum_{i \in I_\lambda} \frac{w(s_i)}{mr} \le \sum_{i \in I_\lambda} \mu(\bar{s}_i) = \int_0^{+\infty} \#\{i \in I_\lambda \mid \mu(\bar{s}_i) \ge t\} dt \le \int_0^{+\infty} \dim \mathcal{F}_w^{mrt} V_\lambda dt.$$

Multiplying by $\varphi(\lambda/(mr)) > 0$ and summing over λ, the formula for $w(D)$ and the formula of the lemma imply $w(D) \le S_{\varphi,m}(v_0; w)$. □

Definition 2.18. Notation as in Set-up 2.10. For $v \in \mathrm{Val}_X$, the *associated graded ring* of v is

$$\mathrm{gr}_v R := \bigoplus_{m \in \mathbb{N}} \bigoplus_{\lambda \in \mathbb{R}} \mathrm{gr}_{\mathcal{F}_v}^{\lambda} R_m,$$

whose multiplication is induced by the multiplication of R [BLXZ23, Section 2.1 and Question 5.3].

3. Quasi-monomiality

In this section, we prove that any valuation computing $\delta_{g,T}(X, \Delta, v_0)$ is quasi-monomial (Theorem 3.1). Thus [LXZ22, Theorem 4.2], which is formulated for quasi-monomial valuations, can be applied to such valuations.

Theorem 3.1. *Notation as in Set-up 2.10. Then any valuation $v \in \mathrm{Val}_X^{T,\circ}$ computing $\delta_{g,T}(X, \Delta, v_0)$ is quasi-monomial.*

We prove Theorem 3.1 by considering the log canonical threshold of the graded sequence $\mathfrak{a}_\bullet(v)$ of valuation ideals of v (cf. [BJ20, Proposition 4.8(ii)]).

Lemma 3.2 extends to $S_g(v_0; \cdot)$ the strict monotonicity of S with respect to the partial order on valuations of Definition 2.3 [BJ20, Proposition 3.15].

Lemma 3.2. *Notation as in Set-up 2.10. Let $u, w \in \mathrm{Val}_X^\circ$ such that $u \le w$. Then*

$$S_g(v_0; w) - S_g(v_0; u) \ge \frac{c_1}{c_2}\big(S(w) - S(u)\big) \ge 0.$$

Moreover, if $u \ne w$, then $S_g(v_0; u) < S_g(v_0; w)$.

Proof. Let m be a positive integer such that $R_m \ne 0$, and let $s \in R_m \backslash \{0\}$. Since $u \le w$, the closure of $\mathrm{center}_X u$ contains $\mathrm{center}_X w$ (Definition 2.3), so any open neighborhood of $\mathrm{center}_X w$ contains $\mathrm{center}_X u$. Thus a trivialization of $\mathcal{O}_X(mL)$ near $\mathrm{center}_X w$ is also a trivialization near $\mathrm{center}_X u$, and the local function f of s under the trivialization belongs to $\mathcal{O}_{X,\mathrm{center}_X w} \subset \mathcal{O}_{X,\mathrm{center}_X u}$. Hence $0 \le u(f) \le w(f)$, that is, $0 \le u(s) \le w(s)$. Therefore $\mathcal{F}_u^0 R_m = R_m$ and $\mathcal{F}_u^t R_m \subset \mathcal{F}_w^t R_m$ for any $t \in \mathbb{R}$. In particular, the image of $\mathcal{F}_{v_0}^{\lambda} R_m \cap \mathcal{F}_u^t R_m$ in $\mathrm{gr}_{\mathcal{F}_{v_0}}^{\lambda} R_m$ is contained in the image of $\mathcal{F}_{v_0}^{\lambda} R_m \cap \mathcal{F}_w^t R_m$, so

$$\dim \mathcal{F}_u^t \mathrm{gr}_{\mathcal{F}_{v_0}}^{\lambda} R_m \le \dim \mathcal{F}_w^t \mathrm{gr}_{\mathcal{F}_{v_0}}^{\lambda} R_m \quad \text{for any } \lambda, t \in \mathbb{R}. \tag{3.1}$$

Since $S_{1,m}(v_0; u) = S_m(u)$ (for a basis $(s_1, \dots, s_{N_m})$ of R_m compatible with $\mathcal{F}_{v_0}$ and $\mathcal{F}_u$, both are equal to $\frac{1}{mrN_m} \sum_{i=1}^{N_m} u(s_i)$ by Definition-Propositions 2.12, 2.14, and 2.15), Lemma 2.17 applied to the constant function $\varphi \equiv 1$, for which $Q_{1,m} = N_m$, implies that

$$S_m(u) = \frac{1}{N_m} \sum_{\lambda \in \mathbb{R}} \int_0^{+\infty} \dim \mathcal{F}_u^{mrt} \mathrm{gr}_{\mathcal{F}_{v_0}}^{\lambda} R_m dt, \tag{3.2}$$

and similarly with u replaced by w.

If $\mathrm{gr}^{\lambda}_{\mathcal{F}_{v_0}} R_m \neq 0$, then $\lambda = v_0(s)$ for some $s \in R_m \backslash \{0\}$, so $0 \leq \lambda \leq mr\lambda_{\max}(\mathcal{F}_{v_0})$ and $g(\frac{\lambda}{mr}) \geq c_1$. For a basis $(s_1, \dots, s_{N_m})$ of R_m compatible with $\mathcal{F}_{v_0}$, we have $0 \leq v_0(s_i) \leq mr\lambda_{\max}(\mathcal{F}_{v_0})$ for any i, so $Q_{g,m} \leq c_2 N_m$. By Lemma 2.17 applied to $\varphi = g$, (3.1), and (3.2),

$$\begin{aligned} S_{g,m}(v_0; w) - S_{g,m}(v_0; u) &= \frac{1}{Q_{g,m}} \sum_{\lambda \in \mathbb{R}} g\Big(\frac{\lambda}{mr}\Big) \int_0^{+\infty} \Big(\dim \mathcal{F}_w^{mrt} \mathrm{gr}^{\lambda}_{\mathcal{F}_{v_0}} R_m \\ &\qquad - \dim \mathcal{F}_u^{mrt} \mathrm{gr}^{\lambda}_{\mathcal{F}_{v_0}} R_m \Big) dt \\ &\geq \frac{c_1}{c_2 N_m} \sum_{\lambda \in \mathbb{R}} \int_0^{+\infty} \Big(\dim \mathcal{F}_w^{mrt} \mathrm{gr}^{\lambda}_{\mathcal{F}_{v_0}} R_m \\ &\qquad - \dim \mathcal{F}_u^{mrt} \mathrm{gr}^{\lambda}_{\mathcal{F}_{v_0}} R_m \Big) dt \\ &= \frac{c_1}{c_2} \big(S_m(w) - S_m(u) \big) \geq 0. \end{aligned}$$

Since L is ample, $R_m \neq 0$ for any $m \gg 0$. Letting $m \to +\infty$, Definition-Propositions 2.12 and 2.15 imply the inequality of the lemma.

Suppose that $u \neq w$. By Definition-Proposition 2.12, the filtrations $\mathcal{F}_u$ and $\mathcal{F}_w$ are linearly bounded, that is, u and w have linear growth in the sense of [BJ20, Section 3.1]. Since L is ample, [BJ20, Proposition 3.15] implies that $S(u) < S(w)$. Since g is continuous and positive, $c_1 > 0$. Thus

$$S_g(v_0; w) - S_g(v_0; u) \geq \frac{c_1}{c_2} \big(S(w) - S(u) \big) > 0,$$

that is, $S_g(v_0; u) < S_g(v_0; w)$. □

Since the infimum defining $\delta_{g,T}(X, \Delta, v_0)$ is taken over T-invariant valuations, while the quasi-monomial valuation w computing $\mathrm{lct}(X, \Delta; \mathfrak{a}_\bullet(v))$ given by [Xu20, Theorem 1.1] need not be T-invariant, in the proof of Theorem 3.1 we replace w by the valuation w^T of Definition-Lemma 3.3.

Definition-Lemma 3.3. Notation as in Set-up 2.10. For a non-zero $f \in \mathcal{O}_X(U)$ on a T-invariant affine open subset $U \subset X$, write $f = \sum_{\chi \in M} f_\chi$, where $f_\chi \in \mathcal{O}_X(U)$ is zero or a T-eigenfunction of weight χ, and $f_\chi = 0$ for all but finitely many χ ($\mathcal{O}_X(U)$ is a union of finite-dimensional T-stable subspaces [Bor91, I.1.9], each of which is a direct sum of weight spaces [Bor91, Proposition 8.4]). Let $w \in \mathrm{Val}_X$. Then there exists a unique valuation $w^T \in \mathrm{Val}_X$ such that

$$w^T(f) = \min\{w(f_\chi) \mid \chi \in M,\ f_\chi \neq 0\}$$

for any T-invariant affine open subset $U \subset X$ containing $\mathrm{center}_X w$ and any non-zero $f \in \mathcal{O}_X(U)$. Moreover, w^T is T-invariant, $w^T(\mathfrak{a}) = w(\mathfrak{a})$ for any non-zero T-invariant ideal sheaf $\mathfrak{a} \subset \mathcal{O}_X$, and $A_{X,\Delta}(w^T) \leq A_{X,\Delta}(w)$. If w is quasi-monomial, then w^T is quasi-monomial.

Proof. The proof is long, so we divide it into several steps. Let $\xi := \mathrm{center}_X w$ and $l := \dim T$. For $t \in T$ and a rational function f on X, let $t^* f := f \circ t$, so that $t^* f = t^{-1} \cdot f$ in the notation of Definition 2.6.

Step 1. In this step we show that there exists a T-invariant affine open subset $U \subset X$ containing ξ.

Since L is ample, there exists a positive integer m divisible by q such that mL is very ample. By Definition-Proposition 2.13(2), R_m is spanned by T-eigensections. Since mL is base point free, there exists a T-eigensection $s \in R_m$ which does not vanish at ξ. Let $U := X \backslash \mathrm{Supp}(\mathrm{div}(s))$. Then U is affine because mL is very ample, and $\xi \in U$. Since s is a T-eigensection, Definition-Proposition 2.13(1) implies that

$$t(\mathrm{div}(s)) = \mathrm{div}(t \cdot s) = \mathrm{div}(s)$$

for any $t \in T$, so U is T-invariant.

Step 2. In this step we construct w^T and show that it is unique and T-invariant.

Fix an isomorphism $T \cong \mathbb{G}_m^l$ with coordinates $t_1, \dots, t_l$, and for $\chi \in M$ denote by t^χ the corresponding monomial in $t_1, \dots, t_l$. Let $\sigma : T \times X \to X$ be the action and $p_2 : T \times X \to X$ the second projection. For a T-invariant affine open subset $U' \subset X$, σ maps $T \times U'$ to U'. By Definition 2.6, $t^* f_\chi = \chi(t)^{-1} f_\chi$ for any $t \in T$, so

$$\sigma^* f = \sum_{\chi \in M} t^{-\chi} p_2^* f_\chi \quad \text{on } T \times U' \text{ for any } f \in \mathcal{O}_X(U'). \tag{3.3}$$

Any rational function on $T \times X$ is a quotient of two non-zero functions of the form $\sum_{\chi \in M} t^\chi p_2^* h_\chi$, where the h_χ are rational functions on X, only finitely many of them non-zero. We call the valuation $\widetilde{w}$ of the field of rational functions on $T \times X$ given by

$$\widetilde{w}\Big(\sum_{\chi \in M} t^\chi p_2^* h_\chi \Big) := \min_{\chi \in M,\ h_\chi \neq 0} w(h_\chi)$$

the *Gauss extension* of w. By [Bou72, Chapter VI, §10, no. 1, Proposition 2] and induction on l (we add the coordinates $t_1, \dots, t_l$ one at a time, and each step is a Gauss extension in one variable), $\widetilde{w}$ is a well-defined valuation, the value group of $\widetilde{w}$ is equal to the value group of w, and the residue field of the valuation ring of $\widetilde{w}$ is generated over the residue field κ_w of the valuation ring of w by l elements algebraically independent over κ_w. Moreover, $\widetilde{w}(c) = 0$ for any $c \in \mathbb{C}^\times$.

Since σ is surjective, σ^* is injective on rational functions. Thus $w^T(f) := \widetilde{w}(\sigma^* f)$ for non-zero rational functions f on X defines a valuation $w^T \in \mathrm{Val}_X$. By (3.3), $w^T(f) = \min_{f_\chi \neq 0} w(f_\chi)$ for any T-invariant affine open subset $U' \subset X$ and any non-zero $f \in \mathcal{O}_X(U')$, in particular when U' contains ξ. A valuation in Val_X is determined by its values on $\mathcal{O}_X(U)$, where U is the open subset of **Step 1**, so w^T is unique. Moreover, for $f \in \mathcal{O}_X(U)$, we have $f_\chi \in \mathcal{O}_X(U) \subset \mathcal{O}_{X,\xi}$, so $w(f_\chi) \geq 0$ for any χ such that $f_\chi \neq 0$, and $w^T(f) \geq 0$. Hence $w^T \geq 0$ on $\mathcal{O}_X(U)$, and $\mathrm{center}_X(w^T) \in U$.

Let $t_0 \in T$ and let $f \in \mathcal{O}_X(U)$ be non-zero. Then $t_0^* f = \sum_{\chi \in M} \chi(t_0)^{-1} f_\chi$, and $\chi(t_0)^{-1} f_\chi$ is zero or a T-eigenfunction of weight χ, so by Definition 2.6,

$$(t_0 \cdot w^T)(f) = w^T(t_0^* f) = \min_{f_\chi \neq 0} w(\chi(t_0)^{-1} f_\chi) = \min_{f_\chi \neq 0} w(f_\chi) = w^T(f).$$

Thus $t_0 \cdot w^T$ and w^T coincide on $\mathcal{O}_X(U)$, so $t_0 \cdot w^T = w^T$, and w^T is T-invariant.

Step 3. In this step we show that $w^T(\mathfrak{a}) = w(\mathfrak{a})$ for any non-zero T-invariant ideal sheaf $\mathfrak{a} \subset \mathcal{O}_X$.

Since U and $\mathfrak{a}$ are T-invariant, $t^* h \in \mathfrak{a}(U)$ for any $h \in \mathfrak{a}(U)$ and $t \in T$. Let $h = \sum_{\chi \in \Phi} h_\chi \in \mathfrak{a}(U)$, where $\Phi \subset M$ is a finite set and $h_\chi \neq 0$ for $\chi \in \Phi$. Since distinct characters are linearly independent functions on T, there exist $\tau_1, \dots, \tau_k \in T$, where $k := \#\Phi$, such that the matrix $(\chi(\tau_i)^{-1})_{1 \leq i \leq k, \chi \in \Phi}$ is invertible. Thus any h_χ is a linear combination of $\tau_i^* h = \sum_{\chi \in \Phi} \chi(\tau_i)^{-1} h_\chi$, $1 \leq i \leq k$, and hence $h_\chi \in \mathfrak{a}(U)$. Since $\mathcal{O}_X(U)$ is Noetherian, $\mathfrak{a}(U)$ is generated by finitely many T-eigenfunctions $h_1, \dots, h_n$. Let $w' \in \{w, w^T\}$ and $\zeta := \mathrm{center}_X w'$. By **Step 2**, $\zeta \in U$, so $\mathfrak{a} \cdot \mathcal{O}_{X,\zeta}$ is generated by $h_1, \dots, h_n$, and $w' \geq 0$ on $\mathcal{O}_{X,\zeta}$. Hence $w'(\mathfrak{a}) = \min_j w'(h_j)$. Since $w^T(h_j) = w(h_j)$ for any j, we have $w^T(\mathfrak{a}) = w(\mathfrak{a})$.

Step 4. In this step we show that $A_{X,\Delta}(w^T) \leq A_{X,\Delta}(w)$.

We first show that for any non-zero $f_1, \dots, f_n \in \mathcal{O}_X(U)$, there exists a non-empty Zariski open subset $T_0 \subset T$ such that $w(t^* f_j) = w^T(f_j)$ for any j and any $t \in T_0$. Since T is irreducible, it suffices to consider one non-zero $f \in \mathcal{O}_X(U)$. Let $\mu := w^T(f)$ and

$$\Phi := \{\chi \in M \mid f_\chi \neq 0,\ w(f_\chi) = \mu\}.$$

Then Φ is a non-empty finite set. Let

$$V := \{h \in \mathcal{O}_{X,\xi} \mid w(h) \geq \mu\} / \{h \in \mathcal{O}_{X,\xi} \mid w(h) > \mu\},$$

and denote by $\bar{h} \in V$ the image of h. For $t \in T$, we have

$$t^*f = \sum_{\chi\in\Phi} \chi(t)^{-1} f_\chi + \sum_{\chi\notin\Phi} \chi(t)^{-1} f_\chi \quad \text{and} \quad w\Big(\sum_{\chi\notin\Phi} \chi(t)^{-1} f_\chi\Big) > \mu,$$

so $w(t^*f) = \mu$ if and only if $\sum_{\chi\in\Phi} \chi(t)^{-1}\bar{f}_\chi \neq 0$. Let $e_1, \dots, e_k$ be a basis of the linear span of the $\bar{f}_\chi$, $\chi \in \Phi$, and write $\bar{f}_\chi = \sum_{i=1}^k a_{\chi,i} e_i$ with $a_{\chi,i} \in \mathbb{C}$. Then

$$\sum_{\chi\in\Phi} \chi(t)^{-1}\bar{f}_\chi = \sum_{i=1}^k P_i(t) e_i, \quad \text{where } P_i := \sum_{\chi\in\Phi} a_{\chi,i} t^{-\chi}.$$

Since $\bar{f}_\chi \neq 0$ for $\chi \in \Phi$, some $a_{\chi,i}$ is non-zero, so some P_i is a non-zero Laurent polynomial, and we may let $T_0 := \{t \in T \mid P_i(t) \neq 0\}$.

Next, we show that w^T belongs to the closure of $\{t \cdot w \mid t \in T\}$ in Val_X. Let $h_1, \dots, h_n$ be non-zero rational functions on X and $\epsilon > 0$. Write $h_j = f_j/f'_j$, where $f_j, f'_j \in \mathcal{O}_X(U)$ are non-zero. By the first paragraph of **Step 4**, there exists a non-empty Zariski open subset $T_0 \subset T$ such that $w(t^*f_j) = w^T(f_j)$ and $w(t^*f'_j) = w^T(f'_j)$ for any j and any $t \in T_0$. Let $t \in T_0$. Then

$$(t \cdot w)(h_j) = w(t^*f_j) - w(t^*f'_j) = w^T(f_j) - w^T(f'_j) = w^T(h_j)$$

for any j, so $|(t \cdot w)(h_j) - w^T(h_j)| < \epsilon$. Thus w^T belongs to the closure of $\{t \cdot w \mid t \in T\}$.

The log discrepancy $A_{X,\Delta}$ is the supremum of the functions $A_{X,\Delta} \circ r_{Y,E}$ over the log smooth models $\pi : (Y, E) \to (X, \Delta)$ with $\mathrm{Supp}(\mathrm{Exc}(\pi) + \pi_*^{-1}\Delta) \subset \mathrm{Supp}\, E$ [LXZ22, Definition 2.5(3)], where $r_{Y,E}$ is the retraction $\mathrm{Val}_X \to \mathrm{QM}(Y, E)$ of [JM12, Section 4.3]. The set $\mathrm{QM}(Y, E)$ is the union of the finitely many cones $\mathrm{QM}_\eta(Y, E)$, each of which is closed in $\mathrm{QM}(Y, E)$ and homeomorphic to a closed simplicial cone [JM12, Lemma 4.5 and Section 4.2]. Since $A_{X,\Delta}$ is linear on each $\mathrm{QM}_\eta(Y, E)$, it is continuous on $\mathrm{QM}(Y, E)$. Since $r_{Y,E}$ is continuous [JM12, Section 4.3], each $A_{X,\Delta} \circ r_{Y,E}$ is continuous. Thus $A_{X,\Delta}$ is lower semicontinuous on Val_X (cf. [JM12, Lemma 5.7]). Moreover, any $t \in T$ is an automorphism of (X, Δ), so $A_{X,\Delta}(t \cdot w) = A_{X,\Delta}(w)$. Suppose that $A_{X,\Delta}(w^T) > A_{X,\Delta}(w)$. Then

$$\{w' \in \mathrm{Val}_X \mid A_{X,\Delta}(w') > A_{X,\Delta}(w)\}$$

is an open subset of Val_X containing w^T, so it contains $t \cdot w$ for some $t \in T$, which contradicts $A_{X,\Delta}(t \cdot w) = A_{X,\Delta}(w)$. Hence $A_{X,\Delta}(w^T) \leq A_{X,\Delta}(w)$.

Step 5. In this step we show that w^T is quasi-monomial if w is quasi-monomial.

Assume that w is quasi-monomial. If w^T is the trivial valuation, then $w^T = v_\alpha$ with $\alpha = 0$ in $\mathrm{QM}_\eta(Y, E)$ for any log smooth model (Y, E) of (X, Δ) and any generic point η of a stratum of (Y, E), so w^T is quasi-monomial. Thus we may assume that w^T is non-trivial. Since w is quasi-monomial, w is Abhyankar [Xu20, Section 2.1.1], that is, $\mathrm{rat.rk}(w) + \mathrm{tr.deg}(w) = d$. By **Step 2**,

$$\mathrm{rat.rk}(\widetilde{w}) + \mathrm{tr.deg}(\widetilde{w}) = \mathrm{rat.rk}(w) + \mathrm{tr.deg}(w) + l = d + l.$$

The field of rational functions on $T \times X$ has transcendence degree l over the image of σ^*, and the restriction of $\widetilde{w}$ to the image of σ^* is w^T via σ^*. By the Zariski–Abhyankar inequality [Bou72, Chapter VI, §10, no. 3, Corollary 1] (see also [Xu20, Section 2.1.1, (2.2)]), applied once to $\widetilde{w}$ and w^T, and once to w^T and the trivial valuation of $\mathbb{C}$,

$$d + l \leq \mathrm{rat.rk}(w^T) + \mathrm{tr.deg}(w^T) + l \leq d + l.$$

Thus $\mathrm{rat.rk}(w^T) + \mathrm{tr.deg}(w^T) = d$, that is, w^T is Abhyankar. Since w^T is non-trivial and Abhyankar, it is quasi-monomial in the sense of [LXZ22, Definition 2.4] by [ELS03, Proposition 2.8] (see also [Xu20, Section 2.1.1]). Let X' be a log resolution of (X, Δ). Since $\mathrm{Val}_X = \mathrm{Val}_{X'}$ and X' is smooth, [JM12, Remark 3.4] implies that $w^T \in \mathrm{QM}(Y', E')$ for a log smooth pair (Y', E') over X' in the sense of [JM12, Definition 3.2], that is, Y' is smooth, E' is a reduced divisor on Y' with simple normal crossing support, and the structure morphism $Y' \to X'$ is proper, birational,

and an isomorphism outside $\operatorname{Supp} E'$. Let F' be the reduced divisor on Y' whose support is the preimage of the exceptional locus of $X' \to X$, and let $\rho : Y' \to X$ be the composite morphism. Since $Y' \to X'$ and $X' \to X$ are isomorphisms outside $\operatorname{Supp} E'$ and the exceptional locus of $X' \to X$ respectively, ρ is an isomorphism outside $\operatorname{Supp}(E' + F')$.

By Chow's lemma [Har77, Chapter II, Exercise 4.10] and [Har77, Chapter II, Theorem 7.17], there exists a non-zero ideal sheaf $\mathfrak{b} \subset \mathcal{O}_X$ such that the blow-up of X along $\mathfrak{b}$ factors through ρ. Let $\psi : Z \to X$ be a projective log resolution of (X, Δ) such that $\mathfrak{b} \cdot \mathcal{O}_Z$ is invertible [KM98, Theorem 0.2]. By the universal property of the blow-up [Har77, Chapter II, Proposition 7.14], $\psi = \rho \circ \phi$ for a morphism $\phi : Z \to Y'$. Since Z is projective and Y' is smooth, $\operatorname{Exc}(\phi)$ is a divisor [KM98, Corollary 2.63]. Let G be the reduced divisor on Z with

$$\operatorname{Supp} G = \phi^{-1}(\operatorname{Supp}(E' + F')) \cup \operatorname{Exc}(\phi) \cup \operatorname{Supp} \psi_*^{-1}\Delta,$$

let $\beta : Y \to Z$ be a projective log resolution of (Z, G) [KM98, Theorem 0.2], let $\pi := \psi \circ \beta = \rho \circ \phi \circ \beta : Y \to X$, and let E be the reduced divisor on Y whose support is $(\phi \circ \beta)^{-1}(\operatorname{Supp}(E' + F')) \cup \operatorname{Exc}(\phi \circ \beta)$ (the set $\operatorname{Exc}(\phi \circ \beta)$ is of pure codimension one by [KM98, Corollary 2.63], since Y is projective and Y' is smooth). Since ρ is an isomorphism outside $\operatorname{Supp}(E' + F')$, we have

$$\operatorname{Exc}(\psi) \subset \operatorname{Exc}(\phi) \cup \phi^{-1}(\operatorname{Supp}(E' + F'))$$

and

$$\operatorname{Exc}(\pi) \subset \operatorname{Exc}(\phi \circ \beta) \cup (\phi \circ \beta)^{-1}(\operatorname{Supp}(E' + F')) = \operatorname{Supp} E.$$

Since Z and X are normal, by Zariski's main theorem [Har77, Chapter V, Theorem 5.2], π is a local isomorphism at a point $y \in Y$ if and only if β is a local isomorphism at y and ψ is a local isomorphism at $\beta(y)$. Thus $\operatorname{Exc}(\pi) = \operatorname{Exc}(\beta) \cup \beta^{-1}(\operatorname{Exc}(\psi))$ is a divisor, since $\operatorname{Exc}(\psi)$ is a divisor on the smooth variety Z, and π is projective. The sets $\operatorname{Supp} E$, $\operatorname{Exc}(\pi)$, and $\operatorname{Supp} \pi_*^{-1}\Delta = \operatorname{Supp} \beta_*^{-1}\psi_*^{-1}\Delta$ are contained in the simple normal crossing divisor $\beta^{-1}(\operatorname{Supp} G) \cup \operatorname{Exc}(\beta)$. Thus $\pi : (Y, E) \to (X, \Delta)$ is a log smooth model of (X, Δ). Since $\phi \circ \beta$ and $Y' \to X'$ are isomorphisms outside $\operatorname{Exc}(\phi \circ \beta) \subset \operatorname{Supp} E$ and $\operatorname{Supp} E'$ respectively, and $(\phi \circ \beta)^{-1}(\operatorname{Supp} E') \subset \operatorname{Supp} E$, (Y, E) dominates (Y', E') as log smooth pairs over X'. By [JM12, Lemma 4.6 and Corollary 5.4], $\operatorname{QM}(Y', E') \subset \operatorname{QM}(Y, E)$. Hence $w^T \in \operatorname{QM}(Y, E)$, that is, w^T is quasi-monomial in the sense of Definition 2.4. □

Proof of Theorem 3.1. Let $v \in \operatorname{Val}_X^{T,\circ}$ be a valuation computing $\delta := \delta_{g,T}(X, \Delta, v_0)$. Since (X, Δ) is klt and v is non-trivial, $A_{X,\Delta}(v) > 0$ [Xu20, Section 2.1.2]. By Lemma 2.16, $0 < S_g(v_0; v) < +\infty$. Thus

$$\delta = \frac{A_{X,\Delta}(v)}{S_g(v_0; v)} > 0.$$

Let $\xi := \operatorname{center}_X v$ and $\mathfrak{a}_m := \mathfrak{a}_m(v)$ for any positive integer m. Since v is T-invariant, $\mathfrak{a}_m$ is a T-invariant ideal sheaf. Since v is non-trivial, ξ is not the generic point of X, so there exists f in the maximal ideal of $\mathcal{O}_{X,\xi}$ such that $p := v(f) > 0$. Then $f^{\lceil m/p \rceil} \in \mathfrak{a}_m \cdot \mathcal{O}_{X,\xi}$, so $\mathfrak{a}_m \neq 0$ and

$$m \leq v(\mathfrak{a}_m) \leq p\left\lceil \frac{m}{p} \right\rceil < m + p.$$

Hence $v(\mathfrak{a}_\bullet) = 1$ (see also [BJ20, Section 1.3]). Let $\gamma := \operatorname{lct}(X, \Delta; \mathfrak{a}_\bullet)$. Since $v \in \operatorname{Val}_X^\circ$ and $v(\mathfrak{a}_\bullet) = 1$,

$$\gamma \leq \frac{A_{X,\Delta}(v)}{v(\mathfrak{a}_\bullet)} = A_{X,\Delta}(v) < +\infty.$$

By [Xu20, Theorem 1.1], there exists a quasi-monomial valuation $w \in \operatorname{Val}_X^\circ$ computing γ. Since $cv_\alpha = v_{c\alpha}$ and $A_{X,\Delta}(cw) = cA_{X,\Delta}(w)$ for any real number $c > 0$, possibly replacing w with $w/w(\mathfrak{a}_\bullet)$, we may assume that $w(\mathfrak{a}_\bullet) = 1$ and $A_{X,\Delta}(w) = \gamma$.

Let $u := w^T$ be the valuation of Definition-Lemma 3.3. Since $\mathfrak{a}_m$ is T-invariant, Definition-Lemma 3.3 implies that

$$u(\mathfrak{a}_m) = w(\mathfrak{a}_m) \geq mw(\mathfrak{a}_\bullet) = m \quad \text{for any } m, \quad \text{and} \quad A_{X,\Delta}(u) \leq A_{X,\Delta}(w) = \gamma \leq A_{X,\Delta}(v),$$

and that u is quasi-monomial and T-invariant. Since $u(\mathfrak{a}_1) \geq 1 > 0$, u is non-trivial. Thus $u \in \mathrm{Val}_X^{T,\circ}$.

We show that $v \leq u$. Let $\zeta := \mathrm{center}_X u$. If $\zeta \notin \overline{\{\xi\}}$, then $\mathfrak{a}_1 \cdot \mathcal{O}_{X,\zeta} = \mathcal{O}_{X,\zeta}$ by Definition 2.7, so $u(\mathfrak{a}_1) = u(1) = 0$, a contradiction. Thus $\zeta \in \overline{\{\xi\}}$, and $\mathcal{O}_{X,\zeta} \subset \mathcal{O}_{X,\xi}$. Let $f \in \mathcal{O}_{X,\zeta}$ be non-zero. Then $v(f) \geq 0$ and $u(f) \geq 0$, so we may assume that $v(f) > 0$. For any positive integer k such that $n_k := \lfloor kv(f) \rfloor \geq 1$, we have $v(f^k) \geq n_k$, so $f^k \in \mathfrak{a}_{n_k} \cdot \mathcal{O}_{X,\zeta}$ by Definition 2.7. Hence

$$ku(f) = u(f^k) \geq u(\mathfrak{a}_{n_k}) \geq n_k = \lfloor kv(f) \rfloor.$$

Dividing by k and letting $k \to +\infty$, we obtain $u(f) \geq v(f)$. Thus $v \leq u$.

Since $u \in \mathrm{Val}_X^{T,\circ}$ and $S_g(v_0; u) > 0$ (Lemma 2.16), the definition of δ implies that $A_{X,\Delta}(u) \geq \delta S_g(v_0; u)$. Since $v \leq u$ and $\delta > 0$, Lemma 3.2 implies that $\delta S_g(v_0; u) \geq \delta S_g(v_0; v)$. Since v computes δ, $\delta S_g(v_0; v) = A_{X,\Delta}(v)$. Thus

$$A_{X,\Delta}(v) \geq A_{X,\Delta}(u) \geq \delta S_g(v_0; u) \geq \delta S_g(v_0; v) = A_{X,\Delta}(v),$$

so all the inequalities are equalities. Since $\delta > 0$, $S_g(v_0; u) = S_g(v_0; v)$. By Lemma 3.2, $u = v$. Therefore $v = w^T$ is quasi-monomial. □

4. Special complements

In this section, we prove Proposition 4.1: if $\delta_{g,T}(X, \Delta, v_0) \leq 1$, then any quasi-monomial valuation computing $\delta_{g,T}(X, \Delta, v_0)$ is a monomial lc place of a special $\mathbb{Q}$-complement. Section 5 uses [LXZ22, Theorem 4.2] for such a valuation. The reason for requiring special $\mathbb{Q}$-complements is explained in the remark after Definition 2.8.

Proposition 4.1. *Notation as in Set-up 2.10. Let $v \in \mathrm{Val}_X^{T,\circ}$ be a quasi-monomial valuation computing $\delta_{g,T}(X, \Delta, v_0)$. Assume that $\delta_{g,T}(X, \Delta, v_0) \leq 1$. Then there exist a log smooth model $\pi : (Y, E) \to (X, \Delta)$ and a $\mathbb{Q}$-complement $(X, \Delta + \Gamma)$ of (X, Δ) such that $(X, \Delta + \Gamma)$ is special with respect to (Y, E) and v is a monomial lc place of $(X, \Delta + \Gamma)$ with respect to (Y, E).*

The proof of Proposition 4.1 is given at the end of Subsection 4.3.

4.1. A uniform complement lemma.

In this subsection, we prove Lemma 4.2, which differs from [LXZ22, Lemma 3.1] and [BLXZ23, Lemma 5.5] in that it uses one boundary B instead of a sequence, $\Gamma \geq D$ for a given $\mathbb{Q}$-divisor $D \leq B$ with $D \in I$ instead of $\Gamma \geq \alpha D$ for a given $\mathbb{Q}$-divisor $D \geq 0$ with $D \sim_{\mathbb{Q}} -(K_X + \Delta)$ and a fixed rational number $\alpha > 0$ as in [LXZ22, Lemma 3.1], and the number η depends only on X, Δ, v, and I, not on B. In the proof of Proposition 4.1, D varies with a general member of a linear system, and the general members available depend on the degree. Since η is fixed before the degree, one degree and one general member suffice, whereas a sequence of boundaries indexed by the degree would require one member that is general for all degrees.

Lemma 4.2 (cf. [LXZ22, Lemma 3.1] and [BLXZ23, Lemma 5.5])**.** *Let (X, Δ) be a log Fano pair, v a quasi-monomial valuation on X with $A_{X,\Delta}(v) = 1$, and $I \subset \mathbb{Q}_{\geq 0}$ a finite set. Then there exists a real number $\eta \in (0, 1)$ depending only on X, Δ, v, and I satisfying the following. Let $B \geq D \geq 0$ be $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisors on X such that $D \in I$, $(X, \Delta + B)$ is klt, $-(K_X + \Delta + B)$ is ample, and $v(B) > 1 - \eta$. Then there exists a $\mathbb{Q}$-complement $(X, \Delta + \Gamma)$ of (X, Δ) such that $\Gamma \geq D$ and $v(\Gamma) = 1$.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we fix a rational number $\rho \in (0, 1)$ which controls the rounding of $\mathbb{Q}$-complements.

Let $d := \dim X$ and

$$J := \{\mathrm{mult}_P \Delta + b \mid P \text{ is a prime divisor on } X,\ b \in I \cup \{0\}\}.$$

Then $J \subset \mathbb{Q}_{\geq 0}$ is a finite set. In [LXZ22, Lemma 3.2], the constant depends only on the dimension, the coefficients of the boundary, and the coefficients of the $\mathbb{Q}$-divisor which is rounded up, which follows from [HMX14, Theorems 1.1 and 1.5]. For any subset $J_0 \subset J$, let ρ_{J_0} be a constant given by [LXZ22, Lemma 3.2] for varieties of dimension d, boundaries whose set of non-zero coefficients is J_0, and reduced divisors to be rounded up. Then ρ_{J_0} is a rational number in $(0, 1]$, because [LXZ22, Lemma 3.2] is formulated for pairs, and the boundary of a pair in [LXZ22, Definition 2.1] is a $\mathbb{Q}$-divisor with non-negative coefficients. Since J has finitely many subsets, we may fix a rational number ρ such that $0 < \rho < 1$ and $\rho \leq \rho_{J_0}$ for any $J_0 \subset J$. The number ρ depends only on d, Δ, and I.

Step 2. In this step we approximate v by finitely many divisorial valuations and fix η.

Since v is quasi-monomial, there exist a log smooth model $\pi : (Y, E) \to (X, \Delta)$, the generic point ξ of a stratum of (Y, E), and a vector α with non-negative entries such that $v = v_\alpha$ in the coordinates of Definition 2.4 at ξ. Since $A_{X,\Delta}(v) = 1$, v is non-trivial, and hence $\alpha \neq 0$. Let E^0 be the sum of the components of E containing ξ whose local equations correspond to the zero entries of α. Then $(E - E^0) + \mathrm{Exc}(\pi) + \pi_*^{-1}\Delta$ has simple normal crossing support, so $\pi : (Y, E - E^0) \to (X, \Delta)$ is a log smooth model, and v is the monomial valuation, with the positive entries of α as weights, at the generic point of the stratum of $(Y, E - E^0)$ containing ξ (the face of $\mathrm{QM}_\xi(Y, E)$ on which the weights along E^0 vanish is the cone QM of that stratum, see [JM12, Section 4.2]). Thus, possibly replacing E with $E - E^0$, we may assume that $E^0 = 0$.

Let W be the stratum of (Y, E) whose generic point is the center of v on Y, and let $E_1, \dots, E_k$ be the components of E containing W. Then $v = v_\alpha$ for some $\alpha \in \mathbb{R}^k_{>0}$ in the coordinates of Definition 2.4 at the generic point of W. For any $\alpha' \in \mathbb{R}^k_{>0}$, the center of $v_{\alpha'}$ on Y is the generic point of W, so the center of $v_{\alpha'}$ on X is the center x of v on X.

By [LXZ22, Definition 2.5], the function $\alpha' \to A_{X,\Delta}(v_{\alpha'})$ is linear on $\mathbb{R}^k_{>0}$. For any non-zero $f \in \mathcal{O}_{X,x}$, the function $\alpha' \to v_{\alpha'}(f)$ is concave on $\mathbb{R}^k_{>0}$, as it is the infimum of the linear functions $\alpha' \to \langle \alpha', \mu \rangle$ over the exponents $\mu \in \mathbb{N}^k$ of the monomials appearing in the expansion of f (see [LXZ22, Definition 2.4] and the text following it). For any positive integer m, $v_{\alpha'}(\mathfrak{a}_m(v))$ is the minimum of $v_{\alpha'}(f)$ over finitely many generators f of the stalk of $\mathfrak{a}_m(v)$ at x, so $\alpha' \to v_{\alpha'}(\mathfrak{a}_m(v))$ is concave. Thus $\alpha' \to v_{\alpha'}(\mathfrak{a}_\bullet(v))$ is concave, and it is finite because

$$0 \leq v_{\alpha'}(\mathfrak{a}_\bullet(v)) \leq v_{\alpha'}(\mathfrak{a}_1(v)) < +\infty.$$

Hence

$$\varphi(\alpha') := A_{X,\Delta}(v_{\alpha'}) - v_{\alpha'}(\mathfrak{a}_\bullet(v))$$

is a finite convex function on $\mathbb{R}^k_{>0}$. Since $v_{t\alpha'} = tv_{\alpha'}$ for any real number $t > 0$, we have $\varphi(t\alpha') = t\varphi(\alpha')$. Since $v(\mathfrak{a}_m(v)) \geq m$ for any positive integer m, we have $v(\mathfrak{a}_\bullet(v)) \geq 1 = A_{X,\Delta}(v)$, that is, $\varphi(\alpha) \leq 0$.

Since φ is a finite convex function on $\mathbb{R}^k_{>0}$, it is Lipschitz near α [Roc70, Theorem 10.4]. Thus there exist real numbers $\epsilon_0 > 0$ and $\kappa > 0$ such that any $\alpha' \in \mathbb{R}^k$ with $|\alpha' - \alpha| \leq \epsilon_0$ belongs to $\mathbb{R}^k_{>0}$ and satisfies

$$|\varphi(\alpha') - \varphi(\alpha)| \leq \kappa |\alpha' - \alpha|.$$

Fix a real number ϵ such that $0 < \epsilon \leq \epsilon_0$ and $\kappa\epsilon \leq \rho/2$. By [LX18, Lemma 2.7], there exist vectors $\beta_1, \dots, \beta_k \in \mathbb{Q}^k$, positive integers $e_1, \dots, e_k$, and positive real numbers $t_1, \dots, t_k$ such that

$$e_i\beta_i \in \mathbb{Z}^k, \quad |\beta_i - \alpha| < \frac{\epsilon}{e_i}, \quad \text{and} \quad \alpha = \sum_{i=1}^k t_i\beta_i.$$

Since $e_i \geq 1$, $|\beta_i - \alpha| < \epsilon \leq \epsilon_0$, so $\beta_i \in \mathbb{Q}^k_{>0}$ and $e_i\beta_i \in \mathbb{Z}^k_{>0}$.

Let g_i be the greatest common divisor of the entries of $e_i\beta_i$. The values of $v_{e_i\beta_i}$ on the non-zero monomials in the local equations of $E_1, \dots, E_k$ are the numbers $\langle e_i\beta_i, \mu \rangle$, $\mu \in \mathbb{N}^k$, so the value group of $v_{e_i\beta_i}$ is $g_i\mathbb{Z}$. By [JM12, Remark 3.9], $v_{e_i\beta_i}$ is a positive multiple of ord_{F_i} for some prime

divisor F_i over X. Since the value group of ord_{F_i} is $\mathbb{Z}$, we have $v_{e_i\beta_i} = g_i \mathrm{ord}_{F_i}$. Let $\gamma_i := \frac{e_i}{g_i}\beta_i$. Then $\mathrm{ord}_{F_i} = v_{\gamma_i}$, and by Definition 2.6,

$$a(F_i, X, \Delta) - \mathrm{ord}_{F_i}(\mathfrak{a}_\bullet(v)) = \varphi(\gamma_i) = \frac{e_i}{g_i}\varphi(\beta_i) \leq \frac{e_i}{g_i}\big(\varphi(\alpha) + \kappa|\beta_i - \alpha|\big) \leq \frac{e_i}{g_i}\kappa|\beta_i - \alpha| < \frac{\kappa\epsilon}{g_i} \leq \frac{\rho}{2}.$$

Since $\beta_i = \frac{g_i}{e_i}\gamma_i$, we have $\alpha = \sum_{i=1}^k \frac{t_i g_i}{e_i}\gamma_i$. If $F_i = F_j$ for some $i \neq j$, then $\gamma_i = \gamma_j$, because the entries of γ_i are the values of ord_{F_i} on the local equations of $E_1, \dots, E_k$. Thus, possibly removing repetitions and adding the corresponding coefficients, we obtain distinct prime divisors $F_1, \dots, F_s$ over X, vectors $\gamma_1, \dots, \gamma_s \in \mathbb{R}^k_{>0}$ such that $\mathrm{ord}_{F_i} = v_{\gamma_i}$, and positive real numbers $c_1, \dots, c_s$ such that

$$\alpha = \sum_{i=1}^s c_i\gamma_i \quad \text{and} \quad a(F_i, X, \Delta) - \mathrm{ord}_{F_i}(\mathfrak{a}_\bullet(v)) < \frac{\rho}{2} \ \text{ for any } i. \tag{4.1}$$

Since

$$0 \leq \mathrm{ord}_{F_i}(\mathfrak{a}_\bullet(v)) \leq \mathrm{ord}_{F_i}(\mathfrak{a}_1(v)) < +\infty,$$

we may fix a real number $\eta \in (0,1)$ such that $\eta \,\mathrm{ord}_{F_i}(\mathfrak{a}_\bullet(v)) < \rho/2$ for any i. Then

$$a(F_i, X, \Delta) - (1-\eta)\,\mathrm{ord}_{F_i}(\mathfrak{a}_\bullet(v)) < \rho \quad \text{for any } i. \tag{4.2}$$

The prime divisors $F_1, \dots, F_s$ and the number η depend only on X, Δ, v, and ρ. Thus η depends only on X, Δ, v, and I.

Step 3. In this step we fix B and D and show that $0 < a(F_i, X, \Delta + B) < \rho$ for any i.

Let $B \geq D \geq 0$ be $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisors on X such that $D \in I$, $(X, \Delta+B)$ is klt, $-(K_X+\Delta+B)$ is ample, and $v(B) > 1-\eta$. Let $a_i := a(F_i, X, \Delta + B)$. Since $(X, \Delta + B)$ is klt, $a_i > 0$. For any prime divisor F over X, the pullback of $K_X + \Delta + B$ to a log resolution of $(X, \Delta + B)$ on which F is a divisor is the sum of the pullbacks of $K_X + \Delta$ and B, so Definition 2.1 implies

$$a(F, X, \Delta + B) = a(F, X, \Delta) - \mathrm{ord}_F(B). \tag{4.3}$$

Let n be a positive integer such that nB is Cartier, and let ψ be a local equation of nB near the center x of v. Then $v(\psi) = nv(B) > n(1-\eta)$. For any positive integer l, let $m_l := \lfloor ln(1-\eta) \rfloor$. If $m_l \geq 1$, then $v(\psi^l) > ln(1-\eta) \geq m_l$, so ψ^l belongs to the stalk of $\mathfrak{a}_{m_l}(v)$ at x. Since the center of ord_{F_i} on X is x by **Step 2**,

$$ln\,\mathrm{ord}_{F_i}(B) = \mathrm{ord}_{F_i}(\psi^l) \geq \mathrm{ord}_{F_i}(\mathfrak{a}_{m_l}(v)) \geq m_l\,\mathrm{ord}_{F_i}(\mathfrak{a}_\bullet(v)).$$

Dividing by ln and letting $l \to +\infty$, we obtain

$$\mathrm{ord}_{F_i}(B) \geq (1-\eta)\,\mathrm{ord}_{F_i}(\mathfrak{a}_\bullet(v)). \tag{4.4}$$

By (4.3), (4.4), and (4.2),

$$a_i = a(F_i, X, \Delta) - \mathrm{ord}_{F_i}(B) \leq a(F_i, X, \Delta) - (1-\eta)\,\mathrm{ord}_{F_i}(\mathfrak{a}_\bullet(v)) < \rho.$$

Thus $0 < a_i < \rho < 1$. Moreover, $a_i \in \mathbb{Q}$ since Δ and B are $\mathbb{Q}$-divisors.

Step 4. In this step we construct a projective birational morphism $f : X' \to X$ such that $F_1, \dots, F_s$ are prime divisors on X', and we show that X' is of Fano type and that $(X', \Delta' + D' + (1-\rho)F)$ is $\mathbb{Q}$-complementary, where Δ', D', and F are defined below.

Let $h : X'' \to X$ be a log resolution of $(X, \Delta + B)$ such that any F_i which is exceptional over X is a prime divisor on X'' (such h exists by [KM98, Lemma 2.45] and [KM98, 0.4(10)]), and let $\mathcal{E}$ be the set of the valuations ord_{F_i} such that F_i is exceptional over X. Since $(X, \Delta + B)$ is klt and $a_i < 1$ for any i, [BCHM10, Corollary 1.4.3], applied to the klt pair $(X, \Delta + B)$ and the set $\mathcal{E}$, which consists of valuations of h-exceptional divisors, implies that there exists a projective birational morphism $f : X' \to X$ such that X' is $\mathbb{Q}$-factorial and the exceptional divisors of f correspond to the elements of $\mathcal{E}$ (if $\mathcal{E} = \emptyset$, then f is small, see the second paragraph after [BCHM10, Corollary 1.4.3]). Thus $F_1, \dots, F_s$ are prime divisors on X', and any f-exceptional prime divisor is one of the F_i. Since X is projective, X' is projective.

Let Δ', B', and D' be the strict transforms of Δ, B, and D on X' with the components $F_1,\dots,F_s$ removed. We remove F_i also when F_i is a divisor on X. Let $F:=\sum_{i=1}^s F_i$. For any prime divisor P on X', the coefficient of P in $f^*(K_X+\Delta+B)-K_{X'}$ is $1-a(P,X,\Delta+B)$ by Definition 2.1 and [KM98, Lemma 2.30]. If $P=F_i$, then $1-a(P,X,\Delta+B)=1-a_i$. If P is not one of the F_i, then P is not exceptional over X, and

$$1-a(P,X,\Delta+B)=\mathrm{mult}_{f_*P}(\Delta+B)=\mathrm{mult}_P(\Delta'+B').$$

Thus

$$K_{X'}+\Delta'+B'+\sum_{i=1}^s(1-a_i)F_i=f^*(K_X+\Delta+B). \tag{4.5}$$

Since $0<a_i<1$, $\Delta'+B'+\sum_i(1-a_i)F_i\geq 0$ is a $\mathbb{Q}$-divisor. By (4.5) and [KM98, Lemma 2.30], $a(P,X',\Delta'+B'+\sum_i(1-a_i)F_i)=a(P,X,\Delta+B)>0$ for any prime divisor P over X', so $(X',\Delta'+B'+\sum_i(1-a_i)F_i)$ is klt. Moreover,

$$-\Big(K_{X'}+\Delta'+B'+\sum_{i=1}^s(1-a_i)F_i\Big)=f^*(-(K_X+\Delta+B))$$

is nef and big. By [PS09, Lemma-Definition 2.6], condition (ii), X' is of Fano type.

Let $N\geq 2$ be an integer such that $-N(K_X+\Delta+B)$ is Cartier and $|-N(K_X+\Delta+B)|$ is base point free. Since $f_*\mathcal{O}_{X'}=\mathcal{O}_X$, the linear system $|f^*(-N(K_X+\Delta+B))|$ is the pullback of $|-N(K_X+\Delta+B)|$, so it is base point free. Let H' be a general member of $|f^*(-N(K_X+\Delta+B))|$. By the Bertini theorem [Har77, Chapter III, Corollary 10.9 and Remark 10.9.2] and [KM98, Lemma 5.17], the pair $(X',\Delta'+B'+\sum_i(1-a_i)F_i+\frac{1}{N}H')$ is klt, and

$$K_{X'}+\Delta'+B'+\sum_{i=1}^s(1-a_i)F_i+\frac{1}{N}H'\sim_{\mathbb{Q}}0$$

by (4.5). Thus $(X',\Delta'+B'+\sum_i(1-a_i)F_i)$ is $\mathbb{Q}$-complementary. Since $D\leq B$ and $a_i<\rho$ for any i,

$$\Delta'+D'+(1-\rho)F\leq\Delta'+B'+\sum_{i=1}^s(1-a_i)F_i.$$

By Definition 2.8, a $\mathbb{Q}$-complement of a pair on X' is also a $\mathbb{Q}$-complement of any pair on X' with a smaller boundary. Since X' is $\mathbb{Q}$-factorial, $(X',\Delta'+D'+(1-\rho)F)$ is a pair, and hence it is $\mathbb{Q}$-complementary.

Step 5. In this step we construct Γ and show that $(X,\Delta+\Gamma)$ is a $\mathbb{Q}$-complement of (X,Δ) such that $\Gamma\geq D$ and $a(F_i,X,\Delta+\Gamma)=0$ for any i.

Let J_0 be the set of non-zero coefficients of $\Delta'+D'$. Any prime divisor P on X' with $\mathrm{mult}_P(\Delta'+D')\neq 0$ is not exceptional over X and is not one of the F_i, and

$$\mathrm{mult}_P(\Delta'+D')=\mathrm{mult}_{f_*P}\Delta+\mathrm{mult}_{f_*P}D\in J$$

since $D\in I$. Thus $J_0\subset J$. Since $\rho\leq\rho_{J_0}$ by **Step 1**, $\Delta'+D'+(1-\rho_{J_0})F\leq\Delta'+D'+(1-\rho)F$, so $(X',\Delta'+D'+(1-\rho_{J_0})F)$ is $\mathbb{Q}$-complementary by **Step 4**. Recall that X' is a $\mathbb{Q}$-factorial projective variety of Fano type with $\dim X'=d$, and F is a reduced divisor, which is $\mathbb{Q}$-Cartier since X' is $\mathbb{Q}$-factorial. By [LXZ22, Lemma 3.2] and the choice of ρ_{J_0} in **Step 1**, if $(X',\Delta'+D'+(1-\rho_{J_0})F)$ is $\mathbb{Q}$-complementary, then so is $(X',\Delta'+D'+F)$. Thus there exists a $\mathbb{Q}$-complement $(X',\Delta'+D'+F+\Gamma')$ of $(X',\Delta'+D'+F)$. For any i, $\mathrm{mult}_{F_i}(\Delta'+D'+F+\Gamma')\geq\mathrm{mult}_{F_i}F=1$ since $\Gamma'\geq 0$, and $\mathrm{mult}_{F_i}(\Delta'+D'+F+\Gamma')\leq 1$ since $(X',\Delta'+D'+F+\Gamma')$ is lc. Thus

$$\mathrm{mult}_{F_i}(\Delta'+D'+F+\Gamma')=1\quad\text{for any } i. \tag{4.6}$$

Let

$$\Gamma:=f_*(\Delta'+D'+F+\Gamma')-\Delta.$$

We show that $\Gamma \geq D$. Let P be a prime divisor on X. If P is not one of the F_i, then

$$\operatorname{mult}_P f_*(\Delta' + D' + F + \Gamma') \geq \operatorname{mult}_P f_*(\Delta' + D') = \operatorname{mult}_P(\Delta + D).$$

If $P = F_i$ for some i, then by (4.6), since $(X, \Delta + B)$ is klt and $B \geq D$,

$$\operatorname{mult}_P f_*(\Delta' + D' + F + \Gamma') = 1 > \operatorname{mult}_P(\Delta + B) \geq \operatorname{mult}_P(\Delta + D).$$

Thus $f_*(\Delta' + D' + F + \Gamma') \geq \Delta + D$, that is, $\Gamma \geq D \geq 0$.

Since $K_{X'} + \Delta' + D' + F + \Gamma' \sim_{\mathbb{Q}} 0$, there exist a positive integer N' and a rational function u on X' such that $N'(K_{X'} + \Delta' + D' + F + \Gamma') = \operatorname{div}(u)$. Since f is birational, u is also a rational function on X, and pushing forward by f implies $N'(K_X + \Delta + \Gamma) = \operatorname{div}(u)$. Thus $K_X + \Delta + \Gamma \sim_{\mathbb{Q}} 0$, and

$$f^*(K_X + \Delta + \Gamma) = \frac{1}{N'} \operatorname{div}(u) = K_{X'} + \Delta' + D' + F + \Gamma'.$$

By [KM98, Lemma 2.30], $a(P, X, \Delta + \Gamma) = a(P, X', \Delta' + D' + F + \Gamma') \geq 0$ for any prime divisor P over X, so $(X, \Delta + \Gamma)$ is lc. Thus $(X, \Delta + \Gamma)$ is a $\mathbb{Q}$-complement of (X, Δ). Moreover, by (4.6),

$$a(F_i, X, \Delta + \Gamma) = 1 - \operatorname{mult}_{F_i}(\Delta' + D' + F + \Gamma') = 0 \quad \text{for any } i.$$

Step 6. In this step we show that $v(\Gamma) = 1$.

Recall from **Step 2** that x is the center of v on X, that $v = v_\alpha$ with $\alpha \in \mathbb{R}^k_{>0}$, and that $\operatorname{ord}_{F_i} = v_{\gamma_i}$ and $\alpha = \sum_i c_i \gamma_i$ with $c_i > 0$ by (4.1). Since $K_X + \Delta$ and $K_X + \Delta + \Gamma$ are $\mathbb{Q}$-Cartier, Γ is $\mathbb{Q}$-Cartier. Let n' be a positive integer such that $n'\Gamma$ is Cartier, and let ψ_Γ be a local equation of $n'\Gamma$ near x. Since the center of $v_{\alpha'}$ on X is x for any $\alpha' \in \mathbb{R}^k_{>0}$, we have $v_{\alpha'}(\Gamma) = \frac{1}{n'} v_{\alpha'}(\psi_\Gamma)$. By **Step 2**, the function $\alpha' \to v_{\alpha'}(\Gamma)$ is finite and concave on $\mathbb{R}^k_{>0}$, and it satisfies $v_{t\alpha'}(\Gamma) = t v_{\alpha'}(\Gamma)$ for any real number $t > 0$, while $\alpha' \to A_{X,\Delta}(v_{\alpha'})$ is linear.

We first show that $v(\Gamma) \geq 1$. Let $c := \sum_i c_i$. By concavity and homogeneity,

$$v(\Gamma) = c v_{\sum_i (c_i/c)\gamma_i}(\Gamma) \geq c \sum_{i=1}^{s} \frac{c_i}{c} v_{\gamma_i}(\Gamma) = \sum_{i=1}^{s} c_i \operatorname{ord}_{F_i}(\Gamma).$$

By Definition 2.1, as in (4.3), $a(F_i, X, \Delta + \Gamma) = a(F_i, X, \Delta) - \operatorname{ord}_{F_i}(\Gamma)$. Since $a(F_i, X, \Delta + \Gamma) = 0$ by **Step 5**, we have

$$\operatorname{ord}_{F_i}(\Gamma) = a(F_i, X, \Delta) = A_{X,\Delta}(v_{\gamma_i}).$$

By the linearity of $\alpha' \to A_{X,\Delta}(v_{\alpha'})$,

$$v(\Gamma) \geq \sum_{i=1}^{s} c_i A_{X,\Delta}(v_{\gamma_i}) = A_{X,\Delta}(v_\alpha) = A_{X,\Delta}(v) = 1.$$

We next show that $v(\Gamma) \leq 1$. Let $\alpha' \in \mathbb{Q}^k_{>0}$. By [JM12, Remark 3.9], $v_{\alpha'} = t \operatorname{ord}_P$ for some real number $t > 0$ and some prime divisor P over X. Since $(X, \Delta + \Gamma)$ is lc, $a(P, X, \Delta + \Gamma) = a(P, X, \Delta) - \operatorname{ord}_P(\Gamma) \geq 0$, so

$$v_{\alpha'}(\Gamma) = t \operatorname{ord}_P(\Gamma) \leq t a(P, X, \Delta) = A_{X,\Delta}(v_{\alpha'}).$$

The functions $\alpha' \to v_{\alpha'}(\Gamma)$ and $\alpha' \to A_{X,\Delta}(v_{\alpha'})$ are continuous on $\mathbb{R}^k_{>0}$ [Roc70, Theorem 10.1], and $\mathbb{Q}^k_{>0}$ is dense in $\mathbb{R}^k_{>0}$. Thus $v_{\alpha'}(\Gamma) \leq A_{X,\Delta}(v_{\alpha'})$ for any $\alpha' \in \mathbb{R}^k_{>0}$. Taking $\alpha' = \alpha$, we obtain $v(\Gamma) \leq A_{X,\Delta}(v) = 1$.

Thus $v(\Gamma) = 1$, and $(X, \Delta + \Gamma)$ is a $\mathbb{Q}$-complement of (X, Δ) such that $\Gamma \geq D$ and $v(\Gamma) = 1$. □

4.2. **Equivariant log smooth models.** Lemma 4.3 is used in **Step 1** of the proof of Proposition 4.1 to obtain a T-invariant linear subspace of some R_p whose general members define divisors on X whose strict transforms on Y are at least an ample $\mathbb{Q}$-divisor. The log smooth pair of [JM12, Remark 3.4] carries no T-action, and the equivariant resolution of [Kol07, Proposition 3.9.1] does not take the valuation into account. We therefore first make the valuation monomial on a product of copies of $\mathbb{P}^1$ with a smooth projective variety on which T acts trivially, and then resolve T-equivariantly.

Lemma 4.3. *Notation as in Set-up 2.10. Let $v \in \mathrm{Val}_X^T$ be a non-trivial quasi-monomial valuation. Then there exist a log smooth model $\pi : (Y, E) \to (X, \Delta)$ and a $\mathbb{Q}$-divisor $F \geq 0$ on Y such that T acts on Y, π is T-equivariant, F is T-invariant, $-F$ is π-ample, and $v \in \mathrm{QM}(Y, E)$.*

Proof. The proof is long, so we divide it into several steps. We write the group law of M additively, let $l := \dim T$, which is the rank of M, and we use the action $t \cdot f = f \circ t^{-1}$ of T on the rational functions on X and the T-eigenfunctions of Definition 2.6. A non-zero rational function on X is T-invariant if and only if it is a T-eigenfunction of weight 0. Since v is T-invariant, $v(t \cdot f) = v(f)$ for any $t \in T$ and any non-zero rational function f on X.

Step 1. In this step we choose T-eigenfunctions $\psi_1, \dots, \psi_l$ whose weights form a basis of M, and we show that $\psi_1, \dots, \psi_l$ are algebraically independent over the field of T-invariant rational functions on X and generate the field of rational functions on X over it.

Let k be a positive integer divisible by q such that kL is very ample. By Definition-Proposition 2.13(2), R_k has a basis $s_0, \dots, s_n$ consisting of T-eigensections, of weights $\chi_0, \dots, \chi_n$. By Definition-Proposition 2.13(1), s_j/s_0 is a T-eigenfunction of weight $\chi_j - \chi_0$ for any j. Let $M' \subset M$ be the subgroup generated by $\chi_1 - \chi_0, \dots, \chi_n - \chi_0$. Suppose that $M' \neq M$. Then the subgroup $T' := \bigcap_{\chi \in M'} \ker \chi$ of T is non-trivial, since the character group of T' is $M/M' \neq 0$ (by the duality between diagonalizable groups and finitely generated abelian groups [Bor91, III.8.2(c) and III.8.12]). Since kL is very ample, any rational function on X is a quotient of two polynomials in $s_1/s_0, \dots, s_n/s_0$, so T' acts trivially on the rational functions on X, and hence on X. This contradicts $T \subset \mathrm{Aut}(X, \Delta)$. Thus $M' = M$. Fix a basis $e_1, \dots, e_l$ of M. Since e_i is an integral linear combination of $\chi_1 - \chi_0, \dots, \chi_n - \chi_0$, there exists a product ψ_i of integral powers of $s_1/s_0, \dots, s_n/s_0$ which is a T-eigenfunction of weight e_i. For $\beta = (\beta_1, \dots, \beta_l) \in \mathbb{Z}^l$, we set $\psi^\beta := \psi_1^{\beta_1} \cdots \psi_l^{\beta_l}$, which is a T-eigenfunction of weight $\chi_\beta := \sum_{i=1}^l \beta_i e_i$.

Let $B \subset \mathbb{Z}^l$ be a finite set, let c_β, $\beta \in B$, be T-invariant rational functions on X, and let $f := \sum_{\beta \in B} c_\beta \psi^\beta$. Since $e_1, \dots, e_l$ is a basis of M, the characters χ_β, $\beta \in B$, are pairwise distinct. For $\beta \neq \beta'$ in B, the set $\{t \in T \mid \chi_\beta(t) = \chi_{\beta'}(t)\}$ is a proper closed subset of T, and T is irreducible, so there exists $\tau \in T$ such that the numbers $\chi_\beta(\tau)$, $\beta \in B$, are pairwise distinct. Since

$$\tau^j \cdot f = \sum_{\beta \in B} \chi_\beta(\tau)^j c_\beta \psi^\beta$$

for any $j \geq 0$ and the Vandermonde matrix $(\chi_\beta(\tau)^j)_{0 \leq j < |B|, \beta \in B}$ is invertible, there exist complex numbers $\lambda_{\beta,j}$ such that

$$c_\beta \psi^\beta = \sum_{j=0}^{|B|-1} \lambda_{\beta,j} \tau^j \cdot f \quad \text{for any } \beta \in B. \tag{4.7}$$

In particular, if $B \subset \mathbb{N}^l$ and $f = 0$, then $c_\beta = 0$ for any $\beta \in B$. Thus $\psi_1, \dots, \psi_l$ are algebraically independent over the field of T-invariant rational functions on X.

Any monomial u in $s_1/s_0, \dots, s_n/s_0$ is a T-eigenfunction of weight χ_β for some $\beta \in \mathbb{Z}^l$, so $u = (u\psi^{-\beta})\psi^\beta$, where $u\psi^{-\beta}$ is T-invariant. Thus, multiplying the numerator and the denominator by ψ^{β_0} for some $\beta_0 \in \mathbb{N}^l$, any rational function on X is a quotient of two rational functions of the form $\sum_{\beta \in B} c_\beta \psi^\beta$, where $B \subset \mathbb{N}^l$ is finite and the c_β are T-invariant. In particular, the field of

rational functions on X is generated by $\psi_1, \dots, \psi_l$ over the field of T-invariant rational functions on X.

Let $y_1, \dots, y_e$ be a transcendence basis of the field of T-invariant rational functions on X over $\mathbb{C}$. By the previous two paragraphs, $e + l = d$, and the field of rational functions on X is finitely generated and algebraic, hence finite, over $\mathbb{C}(y_1, \dots, y_e, \psi_1, \dots, \psi_l)$. By the algebraic independence of $\psi_1, \dots, \psi_l$, T-invariant rational functions which are linearly independent over $\mathbb{C}(y_1, \dots, y_e)$ are linearly independent over $\mathbb{C}(y_1, \dots, y_e, \psi_1, \dots, \psi_l)$. Thus the field of T-invariant rational functions on X is a finite extension of $\mathbb{C}(y_1, \dots, y_e)$, and hence it is finitely generated over $\mathbb{C}$, of transcendence degree $d - l$.

Step 2. In this step we compute v on the rational functions of the form $\sum_\beta c_\beta \psi^\beta$.

For $1 \le i \le l$, let $a_i := v(\psi_i)$. Since replacing ψ_i with ψ_i^{-1} and e_i with $-e_i$ keeps the properties of **Step 1**, possibly replacing ψ_i with ψ_i^{-1} and e_i with $-e_i$ for some i, we may assume that $a_i \ge 0$ for any i. Let $B \subset \mathbb{N}^l$ be a non-empty finite set, let c_β, $\beta \in B$, be non-zero T-invariant rational functions on X, and let $f := \sum_{\beta \in B} c_\beta \psi^\beta$. Then $f \neq 0$ by **Step 1**. Since $v(\tau^j \cdot f) = v(f)$ for any j and $v(\lambda) = 0$ for any $\lambda \in \mathbb{C}^\times$, (4.7) implies $v(c_\beta \psi^\beta) \ge v(f)$ for any $\beta \in B$. Since $v(f) \ge \min_{\beta \in B} v(c_\beta \psi^\beta)$, we obtain

$$v(f) = \min_{\beta \in B} \Big(v(c_\beta) + \sum_{i=1}^{l} \beta_i a_i \Big) \tag{4.8}$$

(cf. [BHJ17, Lemma 4.2] and [XZ20, Section 3.1]).

Step 3. In this step we construct a smooth projective variety Z_1 and a reduced divisor D_{Z_1} with simple normal crossing support on Z_1 such that the valuation induced by v on the T-invariant rational functions is monomial on (Z_1, D_{Z_1}).

By **Step 1**, the field of T-invariant rational functions on X is finitely generated over $\mathbb{C}$, of transcendence degree $d - l$, so it is the field of rational functions of a projective variety Z_0' (the projective closure of an affine variety given by [Har77, Chapter I, Theorem 4.4]). Resolving the singularities of Z_0' [Kol07, Theorem 3.36], we obtain a smooth projective variety Z_0 with $\dim Z_0 = d - l$ and a dominant rational map $\rho : X \dashrightarrow Z_0$ ([Har77, Chapter I, Theorem 4.4]) such that the pullbacks under ρ of the rational functions on Z_0 are exactly the T-invariant rational functions on X. Let $\mu \in \mathrm{Val}_{Z_0}$ be the valuation induced by v, that is, $\mu(c) := v(\rho^* c)$ for any non-zero rational function c on Z_0. Since v is quasi-monomial, v is Abhyankar [Xu20, Section 2.1.1]. By **Step 1**, the field of rational functions on X has transcendence degree l over the field of T-invariant rational functions on X. Thus the Zariski–Abhyankar inequality [Bou72, Chapter VI, §10, no. 3, Corollary 1] (see also [Xu20, Section 2.1.1, (2.2)]), applied to the field of rational functions on X over the field of T-invariant rational functions on X and to the field of rational functions on Z_0 over $\mathbb{C}$, implies

$$d = \mathrm{rat.rk}(v) + \mathrm{tr.deg}(v) \le \mathrm{rat.rk}(\mu) + \mathrm{tr.deg}(\mu) + l \le \dim Z_0 + l = d.$$

Thus μ is Abhyankar, and hence, if μ is non-trivial, quasi-monomial [ELS03, Proposition 2.8].

If μ is trivial, then we let $Z_1 := Z_0$, $D_{Z_1} := 0$, η_{Z_1} the generic point of Z_1, and $k_1 := 0$. Suppose that μ is non-trivial. By [JM12, Remark 3.4], there exists a log smooth pair $(\widetilde{Z}, \widetilde{D})$ over Z_0 in the sense of [JM12, Definition 3.2] such that $\mu \in \mathrm{QM}(\widetilde{Z}, \widetilde{D})$, that is, $\widetilde{Z}$ is smooth, $\widetilde{D}$ is a reduced divisor with simple normal crossing support, and there exists a proper birational morphism $\widetilde{Z} \to Z_0$ which is an isomorphism outside $\widetilde{D}$. By Chow's lemma [Har77, Chapter II, Exercise 4.10], there exist a projective variety $Z^\sharp$ and a morphism $Z^\sharp \to \widetilde{Z}$ which is an isomorphism over a dense open subset $U \subset \widetilde{Z}$. Let $Z^\flat := \mathrm{BR}(Z^\sharp)$ [Kol07, Theorem 3.36]. Then $Z^\flat$ is smooth and projective, and $Z^\flat \to Z^\sharp$ is an isomorphism over the preimage of U, which is smooth. Let $\mathfrak{b} \subset \mathcal{O}_{Z^\flat}$ be the product of the ideal sheaves of the preimages of $\widetilde{D}$ and $\widetilde{Z} \setminus U$, let $Z_1 := \mathrm{BP}(Z^\flat, \mathfrak{b}, \emptyset)$ [Kol07, Theorem 3.35], and let D_{Z_1} be the reduced divisor whose support is the cosupport of the pullback

of $\mathfrak{b}$. Then Z_1 is smooth and projective, D_{Z_1} has simple normal crossing support, $Z_1 \to \widetilde{Z}$ is an isomorphism outside D_{Z_1}, and the preimage of $\widetilde{D}$ is contained in D_{Z_1}. Thus (Z_1, D_{Z_1}) is a log smooth pair over Z_0 which dominates $(\widetilde{Z}, \widetilde{D})$, and hence $\mu \in \mathrm{QM}(Z_1, D_{Z_1})$ [JM12, Lemma 4.6 and Corollary 5.4]. Let η_{Z_1} be the generic point of a stratum of (Z_1, D_{Z_1}) such that $\mu \in \mathrm{QM}_{\eta_{Z_1}}(Z_1, D_{Z_1})$, let $z_1, \dots, z_{k_1} \in \mathcal{O}_{Z_1,\eta_{Z_1}}$ be local equations of the irreducible components of D_{Z_1} containing η_{Z_1}, fix an isomorphism

$$\widehat{\mathcal{O}}_{Z_1,\eta_{Z_1}} \cong \kappa(\eta_{Z_1})[[z_1, \dots, z_{k_1}]],$$

and let $\gamma \in \mathbb{R}^{k_1}_{\geq 0}$ be the vector with $\mu = v_\gamma$.

In both cases, regarding μ as a valuation on Z_1 through the birational map $Z_0 \dashrightarrow Z_1$, we have

$$\mu(c) = \min\{\langle \gamma, \gamma' \rangle \mid c_{\gamma'} \neq 0\} \quad \text{for any non-zero } c = \sum_{\gamma' \in \mathbb{N}^{k_1}} c_{\gamma'} z^{\gamma'} \in \mathcal{O}_{Z_1,\eta_{Z_1}}, \tag{4.9}$$

where $c_{\gamma'} \in \kappa(\eta_{Z_1})$, and where the right-hand side is 0 if $k_1 = 0$. Let $\rho_1 : X \dashrightarrow Z_1$ be the composite of ρ and $Z_0 \dashrightarrow Z_1$.

Step 4. In this step we show that v is monomial on $X' := (\mathbb{P}^1)^l \times Z_1$ with respect to a T-invariant reduced divisor D' with simple normal crossing support.

Let x_i be the coordinate of the i-th factor of $(\mathbb{P}^1)^l$, and let T act on X' by

$$t \cdot (x_1, \dots, x_l, z) := (e_1(t)^{-1} x_1, \dots, e_l(t)^{-1} x_l, z),$$

so that x_i is a T-eigenfunction of weight e_i and T acts trivially on the rational functions pulled back from Z_1. Let $h : X \dashrightarrow X'$ be the rational map with $h^* x_i = \psi_i$ for $1 \leq i \leq l$ whose composite with the second projection $\mathrm{pr}_2 : X' \to Z_1$ is ρ_1. Since ψ_i is a T-eigenfunction of weight e_i and $\rho_1^* c$ is T-invariant for any rational function c on Z_1, h is T-equivariant. The field of rational functions on X' is generated over the field of rational functions on Z_1 by the algebraically independent elements $x_1, \dots, x_l$. Thus, by **Steps 1 and 3**, h^* is an isomorphism from the field of rational functions on X' onto the field of rational functions on X, and h is birational. Through h, we regard v as a valuation on X'. For $1 \leq i \leq l$, let $P_i^0 := \{x_i = 0\}$ and $P_i^\infty := \{x_i = +\infty\}$ be the corresponding prime divisors on X', and let

$$D' := \sum_{i=1}^{l} (P_i^0 + P_i^\infty) + \mathrm{pr}_2^* D_{Z_1}.$$

Then D' is a T-invariant reduced divisor with simple normal crossing support.

Let $I_+ := \{i \mid a_i > 0\}$ and $I_0 := \{i \mid a_i = 0\}$. If $I_+ = \emptyset$ and $k_1 = 0$, then μ is trivial, so (4.8) implies $v(f) = 0$ for any f as in **Step 2**, and hence v is trivial by **Step 1**, a contradiction. Thus $I_+ \neq \emptyset$ or $k_1 \geq 1$. Let η' be the generic point of the irreducible closed subset

$$\bigcap_{i \in I_+} P_i^0 \cap \mathrm{pr}_2^{-1}\left(\overline{\{\eta_{Z_1}\}}\right)$$

of X', which is a stratum of (X', D'). The irreducible components of D' containing η' are P_i^0, $i \in I_+$, and the pullbacks of the irreducible components of D_{Z_1} containing η_{Z_1}, with local equations x_i, $i \in I_+$, and $z_1, \dots, z_{k_1}$. The functions x_i, $i \in I_0$, are units at η', and $\kappa(\eta')$ is the purely transcendental extension of $\kappa(\eta_{Z_1})$ generated by the residues of x_i, $i \in I_0$. Thus any non-zero polynomial in x_i, $i \in I_0$, with coefficients in the coefficient field $\kappa(\eta_{Z_1})$ of $\widehat{\mathcal{O}}_{Z_1,\eta_{Z_1}}$ fixed in **Step 3** is a unit in $\widehat{\mathcal{O}}_{X',\eta'}$, so the quotients P/Q of such polynomials with $Q \neq 0$ form a subfield of $\widehat{\mathcal{O}}_{X',\eta'}$ which maps isomorphically onto $\kappa(\eta')$, and, by the Cohen structure theorem [Eis95, Theorem 7.16 and Proposition 10.16], the isomorphism of **Step 3** extends to an isomorphism

$$\widehat{\mathcal{O}}_{X',\eta'} \cong \kappa(\eta')[[x_i, z_1, \dots, z_{k_1} \mid i \in I_+]]. \tag{4.10}$$

Let $\alpha := ((a_i)_{i \in I_+}, \gamma)$, and let $v_\alpha \in \mathrm{QM}_{\eta'}(X', D')$ be the valuation of Definition 2.4 with respect to (4.10).

We show that $v = v_\alpha$. Any rational function on X' is a quotient of two functions of the form $f = \sum_{\beta\in B} c_\beta x^\beta$, where $B \subset \mathbb{N}^l$ is a non-empty finite set, $x^\beta := x_1^{\beta_1}\cdots x_l^{\beta_l}$, and the c_β are pullbacks of non-zero elements of $\mathcal{O}_{Z_1,\eta_{Z_1}}$. Thus it suffices to show that $v_\alpha(f) = v(f)$ for any such f. Write

$$c_\beta = \sum_{\gamma'\in\mathbb{N}^{k_1}} c_{\beta,\gamma'} z^{\gamma'}$$

with $c_{\beta,\gamma'} \in \kappa(\eta_{Z_1})$, and write $\beta = (\beta^+, \beta^0)$ with $\beta^+ \in \mathbb{N}^{I_+}$ and $\beta^0 \in \mathbb{N}^{I_0}$. Then

$$f = \sum_{\beta^+,\gamma'} \Big(\sum_{\beta^0} c_{(\beta^+,\beta^0),\gamma'} x^{\beta^0}\Big) x^{\beta^+} z^{\gamma'}$$

in $\widehat{\mathcal{O}}_{X',\eta'}$. Since the monomials x^{β^0} are linearly independent over $\kappa(\eta_{Z_1})$, the coefficient of $x^{\beta^+}z^{\gamma'}$, which is an element of $\kappa(\eta')$, is non-zero if and only if $c_{(\beta^+,\beta^0),\gamma'} \neq 0$ for some β^0. Since $a_i = 0$ for $i \in I_0$, $h^*x^\beta = \psi^\beta$, and $\mu(c_\beta) = v(c_\beta)$, it follows from (4.9) and (4.8) that

$$v_\alpha(f) = \min\Big\{\sum_{i=1}^l \beta_i a_i + \langle\gamma,\gamma'\rangle \;\Big|\; c_{\beta,\gamma'} \neq 0\Big\} = \min_{\beta\in B}\Big(\mu(c_\beta) + \sum_{i=1}^l \beta_i a_i\Big) = v(f).$$

Thus $v = v_\alpha \in \mathrm{QM}(X', D')$.

Step 5. In this step we construct $\pi : (Y, E) \to (X, \Delta)$ and show that $v \in \mathrm{QM}(Y, E)$.

Let $X'' \subset X \times X'$ be the closure of the graph of h. Since h is T-equivariant, X'' is invariant under the diagonal action of T, and the projections $\sigma : X'' \to X$ and $\sigma' : X'' \to X'$ are T-equivariant birational morphisms. Let $Y_0 := \mathrm{BR}(X'')$ [Kol07, Theorem 3.36], which is smooth, and let $\Pi_0 : Y_0 \to X''$ be the induced morphism. The action morphism and the second projection $T\times X'' \to X''$ are smooth. Since BR commutes with smooth morphisms [Kol07, Theorem 3.36(4)], the pullbacks of the blow-up sequence Π_0 under both morphisms are $\mathrm{BR}(T \times X'')$. Thus the centers of the blow-ups in Π_0 are T-invariant, and the action of T lifts to Y_0 (cf. [Kol07, Proposition 3.9.1]).

Let $V \subset Y_0$ (resp. $V' \subset Y_0$) be the closed subset of the points at which $\sigma\circ\Pi_0$ (resp. $\sigma'\circ\Pi_0$) is not a local isomorphism. Let $\mathfrak{I} \subset \mathcal{O}_{Y_0}$ be the product of the ideal sheaves of $(\sigma\circ\Pi_0)^{-1}(\operatorname{Supp}\Delta)$, $(\sigma'\circ\Pi_0)^{-1}(\operatorname{Supp} D')$, V, and V', which is a non-zero T-invariant ideal sheaf. Let $Y := \mathrm{BP}(Y_0, \mathfrak{I}, \emptyset)$ [Kol07, Theorem 3.35], and let $\Pi : Y \to Y_0$ be the induced morphism. Then Y is smooth and projective, $\Pi^*\mathfrak{I}$ is the ideal sheaf of a divisor with simple normal crossing support, and Π is an isomorphism over $Y_0\setminus \operatorname{cosupp}\mathfrak{I}$. Since $\mathfrak{I}$ is T-invariant and BP commutes with smooth morphisms [Kol07, Theorem 3.35(4)], the argument of the previous paragraph shows that the centers of the blow-ups in Π are T-invariant and that the action of T lifts to Y. Let E be the reduced divisor whose support is the cosupport of $\Pi^*\mathfrak{I}$, and let

$$\pi := \sigma\circ\Pi_0\circ\Pi : Y \to X \quad\text{and}\quad \pi' := \sigma'\circ\Pi_0\circ\Pi : Y \to X'.$$

Then π and π' are T-equivariant, and E is T-invariant.

Let $y \in Y$. Since Y_0 and X are normal, by Zariski's main theorem [Har77, Chapter V, Theorem 5.2], π is a local isomorphism at y if and only if Π is a local isomorphism at y and $\Pi(y) \notin V$. Thus $\operatorname{Exc}(\pi) = \operatorname{Exc}(\Pi) \cup \Pi^{-1}(V)$. Since $\Pi^*\mathfrak{I}$ is invertible and Y is integral, the pullback of the ideal sheaf of V is invertible, so $\Pi^{-1}(V)$ is a divisor. Since Π is a composite of blow-ups of smooth varieties along smooth centers [Kol07, Theorem 3.35(1)], and any such blow-up is either an isomorphism or has a prime exceptional divisor, $\operatorname{Exc}(\Pi)$ is also a divisor. Both $\operatorname{Exc}(\Pi)$ and $\Pi^{-1}(V)$ are contained in $\operatorname{Supp} E$, as $V \subset \operatorname{cosupp}\mathfrak{I}$. Moreover, $\pi^{-1}(\operatorname{Supp}\Delta) \subset \operatorname{Supp} E$. Thus π is a log resolution of (X, Δ) and $E + \operatorname{Exc}(\pi) + \pi_*^{-1}\Delta$ has simple normal crossing support, that is, $\pi : (Y, E) \to (X, \Delta)$ is a log smooth model. If $y \notin \operatorname{Supp} E$, then Π is a local isomorphism at y and $\Pi(y) \notin V'$, so π' is a local isomorphism at y. Since π' is separated and birational and is a local isomorphism at any point of $Y\setminus\operatorname{Supp} E$, it restricts to an open immersion on $Y\setminus\operatorname{Supp} E$. Moreover, $\pi'^{-1}(\operatorname{Supp} D') \subset \operatorname{Supp} E$. Thus (Y, E) is a log smooth pair over X' in the sense of

[JM12, Definition 3.2] which dominates (X', D'), and hence $v \in \mathrm{QM}(X', D') \subset \mathrm{QM}(Y, E)$ by **Step 4** and [JM12, Lemma 4.6 and Corollary 5.4].

Step 6. In this step we construct F.

By **Step 5**, the morphism $Y \to X''$ is a composite of blow-ups $\pi_j : Y^{(j)} \to Y^{(j-1)}$, $1 \le j \le p$, of T-invariant ideal sheaves $\mathfrak{i}_j \subset \mathcal{O}_{Y^{(j-1)}}$, where $Y^{(0)} = X''$ and $Y^{(p)} = Y$. For any j, let $F_j \ge 0$ be the Cartier divisor on $Y^{(j)}$ with

$$\mathcal{O}_{Y^{(j)}}(-F_j) = \pi_j^{-1}\mathfrak{i}_j \cdot \mathcal{O}_{Y^{(j)}},$$

which is invertible [Har77, Chapter II, Proposition 7.13(a)]. Then F_j is T-invariant. Since $\mathcal{O}_{Y^{(j)}}(-F_j)$ is the sheaf $\mathcal{O}(1)$ of the blow-up π_j, $-F_j$ is π_j-ample [Har77, Chapter II, Proposition 7.10(b)].

Let $H_X := \mathrm{div}(s_0)$, where $s_0 \in R_k$ is the T-eigensection of **Step 1**, and let

$$H_{X'} := \sum_{i=1}^{l} P_i^0 + \mathrm{pr}_2^* H_{Z_1}$$

for an ample divisor $H_{Z_1} \ge 0$ on Z_1. Then H_X and $H_{X'}$ are T-invariant ample Cartier divisors (Definition-Proposition 2.13(1)), so $H^{(0)} := \sigma^* H_X + \sigma'^* H_{X'}$ is a T-invariant ample Cartier divisor on X''. For $1 \le j \le p$, let

$$H^{(j)} := b_j \pi_j^* H^{(j-1)} - F_j,$$

where b_j is a positive integer such that $H^{(j)}$ is ample, which exists by [KM98, Proposition 1.45]. Then $H_Y := H^{(p)}$ is a T-invariant ample Cartier divisor on Y. Since $F_j \ge 0$ for any j, we have $\pi_* H_Y \le b\sigma_* H^{(0)}$, where $b := b_1 \cdots b_p$, and $\sigma_* H^{(0)} \ge 0$.

By Serre's theorem [Har77, Chapter II, Theorem 5.17], applied to the very ample divisor kL of **Step 1** and the ideal sheaf $\mathcal{O}_X(-b\sigma_* H^{(0)})$, there exists a positive integer k' divisible by k such that the linear subspace

$$W := \{s \in R_{k'} \backslash \{0\} \mid \mathrm{div}(s) \ge \pi_* H_Y\} \cup \{0\}$$

of $R_{k'}$ is non-zero. For $t \in T$ and $s \in W \backslash \{0\}$, we have

$$\mathrm{div}(t \cdot s) = t(\mathrm{div}(s)) \ge \pi_* H_Y$$

by Definition-Proposition 2.13(1), as $\pi_* H_Y$ is T-invariant. Thus W is T-invariant, and hence W contains a T-eigensection s' by Definition-Proposition 2.13(2). Let $D := \mathrm{div}(s')$ and $F := \pi^* D - H_Y$. Then D and F are T-invariant, $-F = H_Y - \pi^* D$ is π-ample, and $\pi_* F = D - \pi_* H_Y \ge 0$. By the negativity lemma [KM98, Lemma 3.39], $F \ge 0$. □

4.3. **Construction of special complements.** In this subsection, we construct special $\mathbb{Q}$-complements and prove Proposition 4.1.

The weighted stability threshold $\delta_{g,T}(X, \Delta, v_0)$ only implies $A_{X,\Delta}(w) \ge \delta_{g,T}(X, \Delta, v_0) S_g(v_0; w)$ for $w \in \mathrm{Val}_X^{T,\circ}$, while log canonicity has to be checked on all valuations. The following lemma shows that, for a T-invariant boundary, it suffices to check T-invariant valuations.

Lemma 4.4. *Notation as in Set-up 2.10. Let $D = \sum_i c_i D_i$, where $c_i \ge 0$ are finitely many real numbers and $D_i \ge 0$ are T-invariant $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisors on X. Assume that $u(D) \le A_{X,\Delta}(u)$ for any $u \in \mathrm{Val}_X^{T,\circ}$. Then $(X, \Delta + D)$ is lc.*

Proof. Let P be a prime divisor over X and $w \in \mathrm{Val}_X$ the valuation given by the order of vanishing along P. It suffices to show that $A_{X,\Delta+D}(w) = A_{X,\Delta}(w) - w(D) \ge 0$. For any i, let m_i be a positive integer such that $m_i D_i$ is Cartier. Then the ideal sheaf $\mathcal{O}_X(-m_i D_i)$ is invertible and T-invariant, and $u(D_i) = \frac{1}{m_i} u(\mathcal{O}_X(-m_i D_i))$ for any $u \in \mathrm{Val}_X$. By Definition-Lemma 3.3, $w^T(D_i) = w(D_i)$ for any i, and hence $w^T(D) = w(D)$. If w^T is trivial, then $w(D) = w^T(D) = 0$,

and $A_{X,\Delta}(w) > 0$ as (X, Δ) is klt. If w^T is non-trivial, then $A_{X,\Delta}(w^T) \leq A_{X,\Delta}(w) < +\infty$ by Definition-Lemma 3.3, so $w^T \in \mathrm{Val}_X^{T,\circ}$, and

$$A_{X,\Delta}(w) - w(D) \geq A_{X,\Delta}(w^T) - w^T(D) \geq 0.$$

□

In the proof of Proposition 4.1, the basis defining the unweighted part of the boundary has to be adapted to a filtration other than $\mathcal{F}_{v_0}$ and $\mathcal{F}_v$ (see part (a) of the sketch in Section 1). The following lemma, which splits off the constant $c_1/2$ from g, allows the h-weighted part and the unweighted part to be defined by different bases.

Lemma 4.5. *Notation as in Set-up 2.10. Let $h : \mathbb{R} \to \mathbb{R}_{>0}$ be a continuous function such that $h(x) = g(x) - c_1/2$ for any $x \in [0, \lambda_{\max}(\mathcal{F}_{v_0})]$. Let m be a positive integer such that $R_m \neq 0$, and set $t_m := c_1 N_m/(2Q_{g,m})$. Let D_1 be an h-weighted (m, v_0)-basis type divisor, D_2 an m-basis type divisor, and $D := (1 - t_m)D_1 + t_m D_2$. Then $c_1/(2c_2) \leq t_m < 1$, and*

$$w(D) \leq S_{g,m}(v_0; w)$$

for any $w \in \mathrm{Val}_X^\circ$. Moreover, equality holds if D_1 and D_2 are compatible with $\mathcal{F}_w$.

Proof. For any $s \in R_m \backslash \{0\}$, we have $v_0(s) \geq 0$, and $v_0(s) \leq mr\lambda_{\max}(\mathcal{F}_{v_0})$ by Definition 2.11. Thus $\mathrm{gr}^\lambda_{\mathcal{F}_{v_0}} R_m = 0$ unless $\lambda/(mr) \in [0, \lambda_{\max}(\mathcal{F}_{v_0})]$. Moreover, for a basis $(s_1, \ldots, s_{N_m})$ of R_m compatible with $\mathcal{F}_{v_0}$, we have $g(v_0(s_i)/(mr)) = h(v_0(s_i)/(mr)) + c_1/2$ for any i. Summing over i, we obtain

$$Q_{g,m} = Q_{h,m} + \frac{c_1}{2} N_m. \tag{4.11}$$

Thus $1 - t_m = Q_{h,m}/Q_{g,m} > 0$. Since $Q_{g,m} \leq c_2 N_m$, we have $t_m \geq c_1/(2c_2)$.

Let $w \in \mathrm{Val}_X^\circ$. For $\lambda \in \mathbb{R}$, let

$$\Phi_\lambda := \int_0^{+\infty} \dim \mathcal{F}_w^{mr\tau} \mathrm{gr}^\lambda_{\mathcal{F}_{v_0}} R_m d\tau.$$

Since $g(\lambda/(mr)) = h(\lambda/(mr)) + c_1/2$ whenever $\Phi_\lambda \neq 0$, Lemma 2.17, applied to $\varphi = g$ and to $\varphi = h$, implies

$$Q_{g,m} S_{g,m}(v_0; w) = \sum_{\lambda \in \mathbb{R}} g\Big(\frac{\lambda}{mr}\Big) \Phi_\lambda = Q_{h,m} S_{h,m}(v_0; w) + \frac{c_1}{2} \sum_{\lambda \in \mathbb{R}} \Phi_\lambda.$$

By Lemma 2.17 applied to the constant function $\varphi \equiv 1$, for which $Q_{1,m} = N_m$, we have $N_m S_{1,m}(v_0; w) = \sum_{\lambda \in \mathbb{R}} \Phi_\lambda$. For a basis $(s_1, \ldots, s_{N_m})$ of R_m compatible with $\mathcal{F}_{v_0}$ and $\mathcal{F}_w$, both $S_{1,m}(v_0; w)$ and $S_m(w)$ are equal to $\frac{1}{mrN_m} \sum_{i=1}^{N_m} w(s_i)$ by Definition-Propositions 2.12, 2.14, and 2.15. Thus

$$\sum_{\lambda \in \mathbb{R}} \Phi_\lambda = N_m S_m(w).$$

Dividing by $Q_{g,m}$ and using (4.11), we obtain

$$(1 - t_m) S_{h,m}(v_0; w) + t_m S_m(w) = S_{g,m}(v_0; w). \tag{4.12}$$

By Lemma 2.17 applied to $\varphi = h$, $w(D_1) \leq S_{h,m}(v_0; w)$, and equality holds if D_1 is compatible with $\mathcal{F}_w$. Let $(u_1, \ldots, u_{N_m})$ be the basis of R_m defining D_2. By [BJ20, Lemma 3.5],

$$w(D_2) = \frac{1}{mrN_m} \sum_{i=1}^{N_m} w(u_i) \leq S_m(w),$$

and equality holds if $(u_1, \ldots, u_{N_m})$ is compatible with $\mathcal{F}_w$ by Definition-Proposition 2.12. Since $0 < t_m < 1$, (4.12) implies

$$w(D) = (1 - t_m) w(D_1) + t_m w(D_2) \leq S_{g,m}(v_0; w),$$

and equality holds if D_1 and D_2 are compatible with $\mathcal{F}_w$. □

The idea of the following proof is explained in part (b) of the sketch of the proof in Section 1.

Proof of Proposition 4.1. The proof is long, so we divide it into several steps. Since (X, Δ) is klt and v is non-trivial, $A_{X,\Delta}(v) > 0$ [Xu20, Section 2.1.2]. By Lemma 2.16, $0 < S_g(v_0; v) < +\infty$. Since v computes $\delta_{g,T}(X, \Delta, v_0)$ and $\delta_{g,T}(X, \Delta, v_0) \leq 1$ by assumption,

$$\delta := \delta_{g,T}(X, \Delta, v_0) = \frac{A_{X,\Delta}(v)}{S_g(v_0; v)} \in (0, 1].$$

Step 1. In this step we construct a log smooth model $\pi : (Y, E) \to (X, \Delta)$ with $v \in \mathrm{QM}(Y, E)$ and a T-invariant linear subspace $V \subset R_p$ whose general members s define $\mathbb{Q}$-divisors G_s on X whose strict transforms on Y are at least an ample $\mathbb{Q}$-divisor whose support contains no stratum of (Y, E).

By Lemma 4.3, there exist a log smooth model $\pi : (Y, E) \to (X, \Delta)$ and a $\mathbb{Q}$-divisor $F \geq 0$ on Y such that T acts on Y, π is T-equivariant, F is T-invariant, $-F$ is π-ample, and $v \in \mathrm{QM}(Y, E)$. Since $-(K_X + \Delta)$ is ample, there exists a rational number $\epsilon > 0$ such that

$$A_Y := \pi^*(-(K_X + \Delta)) - \epsilon F$$

is ample [KM98, Proposition 1.45]. Choose a positive integer p such that $q \mid p$, $pr \geq 2$, $pr\epsilon F$ is a divisor with integer coefficients, and the Cartier divisor $prA_Y = \pi^*(pL) - pr\epsilon F$ is very ample. Here pr is an integer because $q \mid p$.

Since X is normal and π is projective and birational, $\pi_*\mathcal{O}_Y = \mathcal{O}_X$ (Zariski's Main Theorem). By the projection formula [Har77, Chapter II, Exercise 5.1(d)],

$$H^0(Y, \mathcal{O}_Y(\pi^*(pL))) = H^0(X, \mathcal{O}_X(pL)) = R_p.$$

Let $V \subset R_p$ be the linear subspace of sections s with $\pi^* \operatorname{div}(s) \geq pr\epsilon F$. Multiplication by the canonical section of $\mathcal{O}_Y(pr\epsilon F)$ is an isomorphism from $H^0(Y, \mathcal{O}_Y(prA_Y))$ onto V. Since π is T-equivariant and F is T-invariant, Definition-Proposition 2.13(1) implies that

$$\pi^* \operatorname{div}(t \cdot s) = t(\pi^* \operatorname{div}(s)) \geq t(pr\epsilon F) = pr\epsilon F$$

for any $t \in T$ and $s \in V$. Thus V is a T-invariant linear subspace of R_p.

For $0 \neq s \in V$, let $H_s := \pi^* \operatorname{div}(s) - pr\epsilon F$, which is the zero divisor of the section of $\mathcal{O}_Y(prA_Y)$ corresponding to s, and let $G_s := \operatorname{div}(s)/(pr)$. Then $G_s \sim_{\mathbb{Q}} -(K_X + \Delta)$ and

$$\pi^* G_s = \epsilon F + \frac{1}{pr} H_s. \tag{4.13}$$

Since prA_Y is very ample, the linear system $|prA_Y| = \{H_s \mid 0 \neq s \in V\}$ is base point free. Hence there exists $s_1 \in V$ such that $\operatorname{center}_Y v \notin \operatorname{Supp} H_{s_1}$. By Definition-Proposition 2.13(2), $s_1 = \sum_\chi s_{1,\chi}$ with $s_{1,\chi} \in V \cap R_{p,\chi}$. If $\operatorname{center}_Y v \in \operatorname{Supp} H_{s_{1,\chi}}$ for any χ with $s_{1,\chi} \neq 0$, then the sections of $\mathcal{O}_Y(prA_Y)$ corresponding to the $s_{1,\chi}$ vanish at $\operatorname{center}_Y v$, and so does the section corresponding to s_1, a contradiction. Thus there exists a T-eigensection $s_0 \in V$ such that $\operatorname{center}_Y v \notin \operatorname{Supp} H_{s_0}$. Then $v(H_{s_0}) = 0$, and (4.13) implies that

$$\alpha := v(s_0) = pr\epsilon v(F) \quad \text{and} \quad v(s) = pr\epsilon v(F) + v(H_s) \geq \alpha \text{ for any } 0 \neq s \in V. \tag{4.14}$$

By Bertini's theorem [Har77, Chapter III, Corollary 10.9], H_s is smooth, hence reduced, for any s in a non-empty Zariski open subset of V. Since $|prA_Y|$ is base point free, for any irreducible closed subset of Y, the sections $s \in V$ such that H_s contains it form a proper linear subspace of V. The set of π-exceptional prime divisors, the set of components of $\pi_*^{-1}\pi_* F$, and the set of strata of (Y, E) are finite. Hence there exists a non-empty Zariski open subset $V^\circ \subset V \backslash \{0\}$ such that for any $s \in V^\circ$, H_s is reduced, H_s contains no π-exceptional prime divisor and no component of $\pi_*^{-1}\pi_* F$, $\operatorname{Supp} H_s$ contains no stratum of (Y, E), and $\operatorname{center}_Y v \notin \operatorname{Supp} H_s$.

Let $s \in V^\circ$. By (4.14), $v(s) = \alpha$. Since H_s is reduced and contains no π-exceptional prime divisor, $\pi_* H_s$ is reduced, and $\pi_* H_s$ has no common component with $\pi_* F$. Applying π_* to (4.13), we obtain $G_s = \epsilon \pi_* F + \pi_* H_s/(pr)$. Thus the coefficient of G_s is $1/(pr)$ along any component of

π_*H_s, $\epsilon\,\mathrm{mult}_P(\pi_*F)$ along any component P of π_*F, and 0 along any other prime divisor on X. Moreover,

$$\pi_*^{-1}G_s = \epsilon\pi_*^{-1}\pi_*F + \frac{1}{pr}H_s \geq \frac{1}{pr}H_s, \tag{4.15}$$

where $H_s/(pr) \sim_{\mathbb{Q}} A_Y$ is ample.

Step 2. In this step we choose a basis (u_i) of R_m compatible with $\mathcal{F}_v$ and with the flag of subspaces of R_m formed by the sections whose divisors are at least $k\,\mathrm{div}(s_0)$, so that replacing the factor $s_0^{k_i}$ of each u_i by s^{k_i} for any $s \in V$ with $v(s) = \alpha$ does not change the values at v.

Let m be a sufficiently large positive integer such that $q \mid m$. For $n \in \mathbb{N}$ and an integer $k \geq 0$, let

$$R_n^{\geq k} := \{s \in R_n \backslash \{0\} \mid \mathrm{div}(s) \geq k\,\mathrm{div}(s_0)\} \cup \{0\}.$$

Since X is normal, a non-zero section $s \in R_n$ belongs to $R_n^{\geq k}$ if and only if $s = s_0^k u'$ for a non-zero section u' of $\mathcal{O}_X((n-kp)L)$, and in this case $kp \leq n$ because L is ample. Thus $R_n^{\geq k} = s_0^k R_{n-kp}$ if $kp \leq n$, and $R_n^{\geq k} = 0$ if $kp > n$. In particular, $\dim R_m^{\geq k} = N_{m-kp}$ for $0 \leq k \leq m/p$. Since s_0 is a T-eigensection, Definition-Proposition 2.13(1) implies that

$$t(\mathrm{div}(s_0)) = \mathrm{div}(t \cdot s_0) = \mathrm{div}(s_0)$$

for any $t \in T$, and hence $R_m^{\geq k}$ is T-invariant. Setting, for any $n \in \mathbb{N}$, the piece of degree n and level λ to be $R_n^{\geq \lceil \lambda \rceil}$ for $\lambda > 0$ and R_n for $\lambda \leq 0$ defines a filtration of R whose pieces in degree m are T-invariant (the multiplicativity in Definition 2.11 follows from the additivity of div on products of sections and the subadditivity of $\lambda \to \lceil \lambda \rceil$, and the other conditions follow from the monotonicity and the left continuity of $\lambda \to \lceil \lambda \rceil$ and from $R_n^{\geq k} = 0$ for $kp > n$). Since $v \in \mathrm{Val}_X^T$, $\mathcal{F}_v^\lambda R_m$ is T-invariant for any λ (Definition-Proposition 2.13(3)). By Definition-Proposition 2.13(4), there exists a basis $(u_1, \ldots, u_{N_m})$ of R_m that consists of T-eigensections and is compatible with $\mathcal{F}_v$ and with the filtration defined by s_0. For any i, let

$$k_i := \max\{k \in \mathbb{Z}_{\geq 0} \mid \mathrm{div}(u_i) \geq k\,\mathrm{div}(s_0)\},$$

and write $u_i = s_0^{k_i} u_i'$ with $0 \neq u_i' \in R_{m-k_ip}$.

For $0 \neq s \in V$, we define

$$D_{2,m}(s) := \frac{1}{mrN_m} \sum_{i=1}^{N_m} \mathrm{div}(s^{k_i} u_i').$$

Since $\mathrm{div}(s) = prG_s$,

$$D_{2,m}(s) = \frac{1}{mrN_m} \sum_{i=1}^{N_m} \mathrm{div}(u_i') + \sigma_m G_s, \quad \text{where} \quad \sigma_m := \frac{p}{mN_m} \sum_{i=1}^{N_m} k_i. \tag{4.16}$$

The first summand does not depend on s, and $D_{2,m}(s_0) = \frac{1}{mrN_m} \sum_i \mathrm{div}(u_i)$ is an m-basis type divisor compatible with $\mathcal{F}_v$.

We bound σ_m. Since $k_ip \leq m$ for any i, $\sigma_m \leq 1$. Let $k := \lfloor m/(2p) \rfloor$. We have $\dim s_0^k R_{m-kp} = N_{m-kp}$, the subspace $s_0^k R_{m-kp}$ is spanned by the u_i that it contains, and $k_i \geq k$ for any such u_i. Thus

$$\sigma_m \geq \frac{pk}{m} \cdot \frac{N_{m-kp}}{N_m}.$$

Since L is an ample Cartier divisor, $N_n = (L^d)n^d/d! + O(n^{d-1})$ [Laz04, Corollary 1.4.41]. Since $pk/m \to 1/2$ and $(m-kp)/m \to 1/2$, the right-hand side tends to $2^{-(d+1)}$ as $m \to +\infty$. Therefore, for any sufficiently large m with $q \mid m$,

$$2^{-(d+2)} \leq \sigma_m \leq 1. \tag{4.17}$$

Let $s \in V$ with $v(s) = \alpha$. Then

$$v(s^{k_i} u_i') = k_i \alpha + v(u_i') = v(s_0^{k_i} u_i') = v(u_i)$$

for any i. Since (u_i) is compatible with $\mathcal{F}_v$, Definition-Proposition 2.12 implies that

$$v(D_{2,m}(s)) = \frac{1}{mrN_m} \sum_{i=1}^{N_m} v(u_i) = S_m(v) = v(D_{2,m}(s_0)). \tag{4.18}$$

Step 3. In this step we combine $D_{2,m}(s)$ with a weighted basis type divisor, prove log canonicity at s_0, and decrease the coefficients of the lc boundary slightly to rational numbers, so that the boundary at s_0 is klt.

Since g is positive and continuous, $c_1 > 0$ and $g(x) - c_1/2 \geq c_1/2$ for any $x \in [0, \lambda_{\max}(\mathcal{F}_{v_0})]$. Hence there exists a continuous function $h : \mathbb{R} \to \mathbb{R}_{>0}$ such that $h(x) = g(x) - c_1/2$ for any $x \in [0, \lambda_{\max}(\mathcal{F}_{v_0})]$, for example the extension of $g - c_1/2$ that is constant on $(-\infty, 0]$ and on $[\lambda_{\max}(\mathcal{F}_{v_0}), +\infty)$. Let $t_m := c_1 N_m/(2Q_{g,m})$. By Lemma 4.5, $c_1/(2c_2) \leq t_m < 1$. Since $v_0, v \in \mathrm{Val}_X^T$, Definition-Proposition 2.13(3)(4) implies that there exists a basis $(y_1, \ldots, y_{N_m})$ of R_m that consists of T-eigensections and is compatible with $\mathcal{F}_{v_0}$ and $\mathcal{F}_v$. Let $D_{1,m}$ be the h-weighted (m, v_0)-basis type divisor of $(y_1, \ldots, y_{N_m})$. For $0 \neq s \in V$, we define

$$D_m(s) := (1 - t_m) D_{1,m} + t_m D_{2,m}(s).$$

Since $D_{1,m}$ and $D_{2,m}(s_0)$ are compatible with $\mathcal{F}_v$, Lemma 4.5 implies that

$$w(D_m(s_0)) \leq S_{g,m}(v_0; w) \text{ for any } w \in \mathrm{Val}_X^\circ, \quad \text{and} \quad v(D_m(s_0)) = S_{g,m}(v_0; v). \tag{4.19}$$

By (4.18), for any $s \in V$ with $v(s) = \alpha$,

$$v(D_m(s)) = (1 - t_m) v(D_{1,m}) + t_m v(D_{2,m}(s_0)) = S_{g,m}(v_0; v). \tag{4.20}$$

For any $w \in \mathrm{Val}_X^\circ$, w has a center on X, so $w(s) \geq 0$ for any $n \in \mathbb{N}$ and any $s \in R_n \backslash \{0\}$, that is, $\mathcal{F}_w^0 R_n = R_n$ for any $n \in \mathbb{N}$, and $\mathcal{F}_w$ is linearly bounded by Definition-Proposition 2.12. Thus by [BLXZ23, Lemma 4.2], there exist a positive integer m_0 and positive real numbers ζ_m with $\lim_{m \to +\infty} \zeta_m = 0$, both independent of w, such that

$$S_{g,m}(v_0; w) \leq (1 + \zeta_m) S_g(v_0; w)$$

for any $w \in \mathrm{Val}_X^\circ$ and any $m \geq m_0$ with $q \mid m$. Let $\mu_m := \delta/(1 + \zeta_m)$. By (4.19) and the definition of δ, for any $w \in \mathrm{Val}_X^{T,\circ}$,

$$\mu_m w(D_m(s_0)) \leq \frac{\delta}{1 + \zeta_m} S_{g,m}(v_0; w) \leq \delta S_g(v_0; w) \leq A_{X,\Delta}(w). \tag{4.21}$$

We write

$$D_m(s) = \sum_j \lambda_{m,j} D_j^{(m)} + \theta_m G_s, \quad \text{where} \quad \theta_m := t_m \sigma_m, \tag{4.22}$$

the sum is finite, $\lambda_{m,j} > 0$ are real numbers, and $D_j^{(m)} = \mathrm{div}(y)/(nr)$ for a T-eigensection $y \in R_n$ with $n \geq 1$ and $q \mid n$. The terms of $D_{1,m}$ are positive multiples of $\mathrm{div}(y)/(mr)$ for $y = y_1, \ldots, y_{N_m}$. By (4.16), the remaining terms are $t_m \,\mathrm{div}(u_i')/(mrN_m)$. For $n_i := m - k_i p \geq 1$,

$$\frac{t_m}{mrN_m} \mathrm{div}(u_i') = \frac{t_m n_i}{mN_m} \cdot \frac{\mathrm{div}(u_i')}{n_i r},$$

and the terms with $n_i = 0$ vanish. Here $q \mid n_i$, and u_i' is a T-eigensection: if $u_i \in R_{m,\chi}$ and $s_0 \in R_{p,\chi_0}$, then Definition-Proposition 2.13(1) implies that

$$(t \cdot s_0)^{k_i} (t \cdot u_i') = t \cdot u_i = \chi(t) s_0^{k_i} u_i',$$

and since R is a domain, $t \cdot u_i' = \chi(t)\chi_0(t)^{-k_i}u_i'$. Since

$$\sum_i h\Big(\frac{v_0(y_i)}{mr}\Big) = Q_{h,m} \quad \text{and} \quad \sum_i (n_i + pk_i) = mN_m,$$

a direct calculation shows that

$$\sum_j \lambda_{m,j} + \theta_m = (1 - t_m) + t_m = 1.$$

By Definition-Proposition 2.13(1), any $D_j^{(m)}$ and G_{s_0} are T-invariant $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisors, and $D_j^{(m)} \sim_{\mathbb{Q}} -(K_X + \Delta)$. By (4.21) and Lemma 4.4, $(X, \Delta + \mu_m D_m(s_0))$ is lc. Moreover, since $S_g(v_0; v) = \lim_{m\to+\infty} S_{g,m}(v_0; v)$,

$$\lim_{m\to+\infty} \mu_m S_{g,m}(v_0; v) = \delta S_g(v_0; v) = A_{X,\Delta}(v).$$

By (4.17) and Lemma 4.5, for any sufficiently large m with $q \mid m$,

$$\beta := \frac{c_1}{2c_2} 2^{-(d+2)} \le \theta_m \le 1.$$

Choose a sequence $e_m \in (0, 1/2)$ with $\lim_{m\to+\infty} e_m = 0$, and choose rational numbers $\lambda'_{m,j}$ and θ'_m such that

$$(1 - e_m)\mu_m\lambda_{m,j} < \lambda'_{m,j} < \Big(1 - \frac{e_m}{2}\Big)\mu_m\lambda_{m,j} \quad \text{and} \quad (1 - e_m)\mu_m\theta_m < \theta'_m < \Big(1 - \frac{e_m}{2}\Big)\mu_m\theta_m.$$

For $0 \ne s \in V$, let

$$B_m(s) := \sum_j \lambda'_{m,j} D_j^{(m)} + \theta'_m G_s \quad \text{and} \quad \tau_m := \sum_j \lambda'_{m,j} + \theta'_m.$$

Then $B_m(s) \ge 0$ is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor,

$$(1 - e_m)\mu_m D_m(s) \le B_m(s) \le \Big(1 - \frac{e_m}{2}\Big)\mu_m D_m(s), \tag{4.23}$$

$B_m(s) \sim_{\mathbb{Q}} \tau_m(-(K_X + \Delta))$, and $0 < \tau_m < \mu_m \le \delta \le 1$. Thus

$$-(K_X + \Delta + B_m(s)) \sim_{\mathbb{Q}} (1 - \tau_m)(-(K_X + \Delta))$$

is ample.

We show that $(X, \Delta + B_m(s_0))$ is klt. Let w be a divisorial valuation on X. Since $A_{X,\Delta+D}(w) = A_{X,\Delta}(w) - w(D)$ for any $\mathbb{R}$-Cartier $\mathbb{R}$-divisor $D \ge 0$ (Definition 2.1), (4.23) implies that

$$\begin{aligned} A_{X,\Delta+B_m(s_0)}(w) &\ge A_{X,\Delta+(1-e_m/2)\mu_m D_m(s_0)}(w) \\ &= \frac{e_m}{2} A_{X,\Delta}(w) + \Big(1 - \frac{e_m}{2}\Big) A_{X,\Delta+\mu_m D_m(s_0)}(w) > 0, \end{aligned}$$

because (X, Δ) is klt and $(X, \Delta + \mu_m D_m(s_0))$ is lc.

Fix a rational number b with $0 < b < \delta\beta/4$. For any sufficiently large m with $q \mid m$, $\zeta_m \le 1$, hence $\mu_m \ge \delta/2$, and

$$\theta'_m > (1 - e_m)\mu_m\theta_m > \frac{1}{2} \cdot \frac{\delta}{2} \cdot \beta > b.$$

Thus

$$B_m(s) \ge \theta'_m G_s \ge bG_s \quad \text{for any } 0 \ne s \in V. \tag{4.24}$$

Let

$$I := \{0\} \cup \Big\{\frac{b}{pr}\Big\} \cup \{b\epsilon \operatorname{mult}_P(\pi_* F) \mid P \text{ is a component of } \pi_* F\} \subset \mathbb{Q}_{\ge 0}.$$

The number b depends only on δ, c_1, c_2, and d, so I is a finite set independent of s and m. By **Step 1**, $bG_s \in I$ for any $s \in V^\circ$. Finally, for any $s \in V$ with $v(s) = \alpha$, (4.20) and (4.23) imply that

$$(1-e_m)\mu_m S_{g,m}(v_0;v) \le v(B_m(s)) \le \Big(1-\frac{e_m}{2}\Big)\mu_m S_{g,m}(v_0;v), \tag{4.25}$$

and both bounds tend to $A_{X,\Delta}(v)$ as $m \to +\infty$.

Step 4. In this step we fix m and show that $(X, \Delta + B_m(s))$ is klt for any s in a non-empty Zariski open subset of V.

Let m be a positive integer such that $q \mid m$ and the conclusions of **Steps 2 and 3** hold. Let $f : W \to Y$ be a log resolution of the closed subset

$$\pi^{-1}\Big(\operatorname{Supp}\Big(\Delta + \sum_j D_j^{(m)}\Big)\Big) \cup \operatorname{Exc}(\pi) \cup \operatorname{Supp} F$$

of Y, and let B_W be the $\mathbb{Q}$-divisor on W defined by

$$K_W + B_W = (\pi \circ f)^*\Big(K_X + \Delta + \sum_j \lambda'_{m,j} D_j^{(m)}\Big) + \theta'_m \epsilon f^* F.$$

Then $\operatorname{Supp} B_W$ is a simple normal crossing divisor, and by (4.13),

$$K_W + B_W + \frac{\theta'_m}{pr} f^* H_s = (\pi \circ f)^*(K_X + \Delta + B_m(s)) \quad \text{for any } 0 \ne s \in V. \tag{4.26}$$

By **Step 3**, $(X, \Delta + B_m(s_0))$ is klt. By (4.26) for $s = s_0$ and [KM98, Lemma 2.30], any coefficient of $B_W + (\theta'_m/(pr))f^* H_{s_0}$ is less than 1. Since $f^* H_{s_0} \ge 0$, any coefficient of B_W is less than 1.

The divisors $f^* H_s$, $0 \ne s \in V$, form a base point free linear system on W. Any stratum Z of $\operatorname{Supp} B_W$ is a non-singular projective variety, and the restrictions of $f^* H_s$, $0 \ne s \in V$, to Z form a base point free linear system on Z. Applying Bertini's theorem [Har77, Chapter III, Corollary 10.9] to W and to any such Z, we obtain a non-empty Zariski open subset of V such that for any s in it, $f^* H_s$ is smooth, $f^* H_s$ has no common component with $\operatorname{Supp} B_W$, and $\operatorname{Supp} B_W + f^* H_s$ is a simple normal crossing divisor. For such s, the coefficients of $B_W + (\theta'_m/(pr))f^* H_s$ along the components of $f^* H_s$ are equal to $\theta'_m/(pr)$, and

$$\frac{\theta'_m}{pr} < \frac{\theta_m}{pr} \le \frac{1}{2} < 1$$

because $\theta'_m < (1 - e_m/2)\mu_m \theta_m < \theta_m \le 1$ by $\mu_m \le \delta \le 1$ (**Step 3**), and $pr \ge 2$. Hence W is smooth, $B_W + (\theta'_m/(pr))f^* H_s$ has simple normal crossing support, and any coefficient of $B_W + (\theta'_m/(pr))f^* H_s$ is less than 1. Since the sum of any two such coefficients is less than 2, [KM98, Corollary 2.31(3)] implies that any prime divisor over W which is exceptional over W has positive log discrepancy with respect to $(W, B_W + (\theta'_m/(pr))f^* H_s)$ (the discrepancy of [KM98, Definition 2.25] is the log discrepancy minus 1, and negative coefficients are allowed there). Since any coefficient of $B_W + (\theta'_m/(pr))f^* H_s$ is less than 1, the sub-pair $(W, B_W + (\theta'_m/(pr))f^* H_s)$ is sub-klt. By (4.26) and [KM98, Lemma 2.30], $(X, \Delta + B_m(s))$ is klt.

Step 5. In this step we show, by Lemma 4.2 with $B = B_m(s)$ for a general $s \in V$, that (X, Δ) has a special $\mathbb{Q}$-complement $(X, \Delta + \Gamma)$ with v as a monomial lc place.

Since v is quasi-monomial, $v/A_{X,\Delta}(v)$ is quasi-monomial and $A_{X,\Delta}(v/A_{X,\Delta}(v)) = 1$ [LXZ22, Definition 2.5]. Recall that I is the finite set defined in **Step 3**, which does not depend on m. By Lemma 4.2 applied to (X, Δ), $v/A_{X,\Delta}(v)$, and I, there exists a real number $\eta \in (0, 1)$ satisfying the conclusion of Lemma 4.2. By **Step 3**, we may choose a positive integer m as in **Step 4** such that

$$\frac{(1-e_m)\mu_m S_{g,m}(v_0;v)}{A_{X,\Delta}(v)} > 1 - \eta.$$

Recall that V° is the non-empty Zariski open subset of V defined in **Step 1**. Let $V_m \subset V$ be the non-empty Zariski open subset given by **Step 4** for m. Since V is irreducible, $V_m \cap V^\circ \neq \emptyset$. Choose $s \in V_m \cap V^\circ$. Then the following hold.

(1) $B_m(s) \geq bG_s \geq 0$ are $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisors and $bG_s \in I$ (**Step 3**).
(2) $(X, \Delta + B_m(s))$ is klt (**Step 4**), and $-(K_X + \Delta + B_m(s))$ is ample (**Step 3**).
(3) Since $s \in V^\circ$, $v(s) = \alpha$, and by (4.25),

$$\frac{v(B_m(s))}{A_{X,\Delta}(v)} \geq \frac{(1-e_m)\mu_m S_{g,m}(v_0; v)}{A_{X,\Delta}(v)} > 1 - \eta.$$

By Lemma 4.2 with $D = bG_s$ and $B = B_m(s)$, there exists a $\mathbb{Q}$-complement $(X, \Delta + \Gamma)$ of (X, Δ) such that $\Gamma \geq bG_s$ and $v(\Gamma) = A_{X,\Delta}(v)$. In particular (cf. [Xu20, Definition 2.1(4)]),

$$A_{X,\Delta+\Gamma}(v) = A_{X,\Delta}(v) - v(\Gamma) = 0.$$

By (4.15),

$$\pi_*^{-1}\Gamma \geq b\pi_*^{-1}G_s \geq \frac{b}{pr}H_s,$$

where $\frac{b}{pr}H_s \geq 0$ is an ample $\mathbb{Q}$-divisor on Y whose support $\operatorname{Supp} H_s$ contains no stratum of (Y, E) because $s \in V^\circ$. Thus $(X, \Delta + \Gamma)$ is special with respect to (Y, E). Since $v \in \mathrm{QM}(Y, E)$ is non-trivial (**Step 1**) and $A_{X,\Delta+\Gamma}(v) = 0$, v is a monomial lc place of $(X, \Delta + \Gamma)$ with respect to (Y, E). □

5. Proof of the main theorem

In this section, we prove Theorem 1.1. Since [LXZ22, Theorem 4.2] requires an integer index, we first apply it to the Veronese subring $R^{(q)} := \bigoplus_{m\in\mathbb{N}} R_{qm}$ of R, where q is as in Definition-Proposition 2.13, and then return to R by Lemma 5.1.

The graded ring $\operatorname{gr}_v R$ is not known to be Noetherian in advance, and integrality over a finitely generated subring does not imply finite generation. In the following lemma, the bound on the degree of the extension of fraction fields and the finiteness of integral closure [Eis95, Corollary 13.13] imply module-finiteness.

Lemma 5.1. *Let $A = \bigoplus_{m\in\mathbb{N}} A_m$ be an $\mathbb{N}$-graded $\mathbb{C}$-algebra that is a domain, q a positive integer, and $A^{(q)} := \bigoplus_{m\in\mathbb{N}} A_{qm}$. Assume that $A^{(q)}$ is a finitely generated $\mathbb{C}$-algebra. Then A is a finitely generated $A^{(q)}$-module. In particular, A is a finitely generated $\mathbb{C}$-algebra.*

Proof. We regard $A^{(q)}$ as a subring of A. For any homogeneous element $a \in A$, we have $a^q \in A^{(q)}$, so a is a root of the monic polynomial $Z^q - a^q$ with coefficients in $A^{(q)}$. Since the elements of A that are integral over $A^{(q)}$ form a subring of A [AM69, Corollary 5.3], and any element of A is a finite sum of homogeneous elements, A is integral over $A^{(q)}$.

We show that

$$[\operatorname{Frac}(A) : \operatorname{Frac}(A^{(q)})] \leq q.$$

Let J be the set of integers $j \in \{0, 1, \dots, q-1\}$ such that A contains a non-zero homogeneous element of degree congruent to j modulo q. For $j \in J$, we pick a non-zero homogeneous element $b_j \in A$ of degree congruent to j modulo q, and we let $b_0 := 1$. Let $a \in A$ be a non-zero homogeneous element of degree congruent to $j \in J$ modulo q. Since ab_j^{q-1} and b_j^q are homogeneous of degrees divisible by q, and $b_j^q \neq 0$, we have

$$\frac{a}{b_j} = \frac{ab_j^{q-1}}{b_j^q} \in \operatorname{Frac}(A^{(q)}).$$

Let $V \subset \operatorname{Frac}(A)$ be the $\operatorname{Frac}(A^{(q)})$-linear span of $\{b_j\}_{j\in J}$. Then $A \subset V$. For $i, j \in J$, the element $b_i b_j$ is non-zero and homogeneous, so $b_i b_j \in \operatorname{Frac}(A^{(q)})b_k$, where $k \in J$ is the residue of $i + j$

modulo q. Thus V is a subring of $\mathrm{Frac}(A)$ containing $\mathrm{Frac}(A^{(q)})$, and

$$\dim_{\mathrm{Frac}(A^{(q)})} V \leq |J| \leq q.$$

For any non-zero $x \in V$, multiplication by x is an injective $\mathrm{Frac}(A^{(q)})$-linear endomorphism of V, hence it is surjective, and x is invertible in V. Thus V is a field. Since $A \subset V \subset \mathrm{Frac}(A)$, we have $V = \mathrm{Frac}(A)$, and $[\mathrm{Frac}(A) : \mathrm{Frac}(A^{(q)})] \leq q$.

Let $\bar{A}$ be the integral closure of $A^{(q)}$ in $\mathrm{Frac}(A)$. Since $A^{(q)}$ is a finitely generated $\mathbb{C}$-algebra and a domain, and $\mathrm{Frac}(A)$ is a finite extension of $\mathrm{Frac}(A^{(q)})$, $\bar{A}$ is a finitely generated $A^{(q)}$-module [Eis95, Corollary 13.13]. Since A is integral over $A^{(q)}$, we have $A \subset \bar{A}$. The ring $A^{(q)}$ is Noetherian [AM69, Corollary 7.7], so $\bar{A}$ is a Noetherian $A^{(q)}$-module [AM69, Proposition 6.5], and the submodule A of $\bar{A}$ is a finitely generated $A^{(q)}$-module [AM69, Proposition 6.2]. Thus A is generated as a $\mathbb{C}$-algebra by finitely many generators of the $\mathbb{C}$-algebra $A^{(q)}$ together with finitely many generators of the $A^{(q)}$-module A. □

Proof of Theorem 1.1. The data in Theorem 1.1 satisfy the conditions of Set-up 2.10. Let $v \in \mathrm{Val}_X^{T,\circ}$ be a valuation computing $\delta_{g,T}(X, \Delta, v_0)$. By Theorem 3.1, v is quasi-monomial. Since $\delta_{g,T}(X, \Delta, v_0) \leq 1$, by Proposition 4.1, there exist a log smooth model $\pi : (Y, E) \to (X, \Delta)$ and a $\mathbb{Q}$-complement $(X, \Delta + \Gamma)$ of (X, Δ) such that $(X, \Delta + \Gamma)$ is special with respect to (Y, E) and v is a monomial lc place of $(X, \Delta + \Gamma)$ with respect to (Y, E).

Recall that qr is a positive integer, $qr(K_X + \Delta) = -qL$ is Cartier, and

$$R^{(q)} = \bigoplus_{m \in \mathbb{N}} R_{qm} = \bigoplus_{m \in \mathbb{N}} H^0(X, \mathcal{O}_X(-mqr(K_X + \Delta))).$$

Since (X, Δ) is klt and $v \in \mathrm{Val}_X^\circ$, we have $0 < A_{X,\Delta}(v) < +\infty$ (see the paragraph after [LXZ22, Definition 2.5]). Since $\mathrm{QM}(Y, E)$ is invariant under multiplication by positive real numbers, and $A_{X,\Delta+\Gamma}$ and $A_{X,\Delta}$ are homogeneous of degree one [LXZ22, Definition 2.5], we have

$$\frac{v}{A_{X,\Delta}(v)} \in \mathrm{QM}(Y, E), \quad A_{X,\Delta+\Gamma}\left(\frac{v}{A_{X,\Delta}(v)}\right) = 0, \quad \text{and} \quad A_{X,\Delta}\left(\frac{v}{A_{X,\Delta}(v)}\right) = 1.$$

Thus $v/A_{X,\Delta}(v)$ is a monomial lc place of the special $\mathbb{Q}$-complement $(X, \Delta + \Gamma)$ of (X, Δ) in the sense of [LXZ22, Definition 3.3]. By [LXZ22, Theorem 4.2], for any monomial lc place w of a special $\mathbb{Q}$-complement $(X, \Delta + \Gamma)$ of (X, Δ) and any positive integer r' such that $r'(K_X + \Delta)$ is Cartier, the ring

$$\mathrm{gr}_w \bigoplus_{m \in \mathbb{N}} H^0(X, \mathcal{O}_X(-mr'(K_X + \Delta)))$$

is finitely generated. In particular, for $w = v/A_{X,\Delta}(v)$ and $r' = qr$, the ring $\mathrm{gr}_{v/A_{X,\Delta}(v)} R^{(q)}$ is finitely generated. For any $m \in \mathbb{N}$ and $\lambda \in \mathbb{R}$, we have

$$\mathcal{F}^\lambda_{v/A_{X,\Delta}(v)} R_{qm} = \mathcal{F}_v^{\lambda A_{X,\Delta}(v)} R_{qm} \quad \text{and} \quad \mathcal{F}^{>\lambda}_{v/A_{X,\Delta}(v)} R_{qm} = \mathcal{F}_v^{>\lambda A_{X,\Delta}(v)} R_{qm}.$$

Thus

$$\mathrm{gr}_{v/A_{X,\Delta}(v)} R^{(q)} = \mathrm{gr}_v R^{(q)}$$

as $\mathbb{C}$-algebras, where the real degree λ on the left corresponds to the real degree $\lambda A_{X,\Delta}(v)$ on the right. Hence $\mathrm{gr}_v R^{(q)}$ is a finitely generated $\mathbb{C}$-algebra.

We grade $\mathrm{gr}_v R$ by the section degree, that is, for $m \in \mathbb{N}$,

$$(\mathrm{gr}_v R)_m := \bigoplus_{\lambda \in \mathbb{R}} \mathrm{gr}^\lambda_{\mathcal{F}_v} R_m.$$

The degree m part of $R^{(q)}$ is R_{qm} with the filtration $\mathcal{F}_v$ of R_{qm}, and the multiplication of $R^{(q)}$ is the restriction of the multiplication of R. Thus

$$\mathrm{gr}_v R^{(q)} = \bigoplus_{m \in \mathbb{N}} \bigoplus_{\lambda \in \mathbb{R}} \mathrm{gr}^\lambda_{\mathcal{F}_v} R_{qm} = (\mathrm{gr}_v R)^{(q)}$$

as $\mathbb{C}$-algebras.

We show that $\operatorname{gr}_v R$ is a domain. For non-zero $s \in R_m$ and $t \in R_{m'}$, computing v with a local trivialization of L at the center of v, we have $v(st) = v(s) + v(t)$. A non-zero element of $\operatorname{gr}_v R$ of bidegree (m, λ) is the class of some $s \in R_m$ with $v(s) = \lambda$, so the product of two non-zero elements of bidegrees (m, λ) and (m', λ') is a non-zero element of bidegree $(m + m', \lambda + \lambda')$. We order $\mathbb{Z} \times \mathbb{R}$ lexicographically, which is a total order compatible with addition. Let $x, y \in \operatorname{gr}_v R$ be non-zero elements, and let x_1 and y_1 be the non-zero homogeneous components of x and y of the greatest bidegrees β and γ respectively. Any other pair of bidegrees of non-zero components of x and y has sum strictly less than $\beta + \gamma$, so the component of xy of bidegree $\beta + \gamma$ is $x_1 y_1 \neq 0$. Thus $xy \neq 0$.

Therefore, $\operatorname{gr}_v R$ is an $\mathbb{N}$-graded $\mathbb{C}$-algebra that is a domain, and $(\operatorname{gr}_v R)^{(q)}$ is a finitely generated $\mathbb{C}$-algebra. By Lemma 5.1, $\operatorname{gr}_v R$ is a finitely generated $\mathbb{C}$-algebra. □

References

[AZ22] H. Abban and Z. Zhuang, *K-stability of Fano varieties via admissible flags*, Forum Math. Pi **10** (2022), Paper No. e15.

[AM69] M. F. Atiyah and I. G. Macdonald, *Introduction to commutative algebra*, Addison-Wesley Publishing Co., Reading, Mass.-London-Don Mills, Ont. (1969).

[BCHM10] C. Birkar, P. Cascini, C. D. Hacon, and J. McKernan, *Existence of minimal models for varieties of log general type*, J. Amer. Math. Soc. **23** (2010), no. 2, 405–468.

[BJ20] H. Blum and M. Jonsson, *Thresholds, valuations, and K-stability*, Adv. Math. **365** (2020), Paper No. 107062, 57 pp.

[BLXZ23] H. Blum, Y. Liu, C. Xu, and Z. Zhuang, *The existence of the Kähler–Ricci soliton degeneration*, Forum Math. Pi **11** (2023), Paper No. e9, 28 pp.

[BLXZ25] H. Blum, Y. Liu, C. Xu, and Z. Zhuang, *Relative stability theory and properness of K-moduli spaces*, arXiv:2510.06197.

[BLZ22] H. Blum, Y. Liu, and C. Zhou, *Optimal destabilization of K-unstable Fano varieties via stability thresholds*, Geom. Topol. **26** (2022), no. 6, 2507–2564.

[Bor91] A. Borel, *Linear algebraic groups*, 2nd ed., Graduate Texts in Mathematics **126**, Springer-Verlag, New York (1991).

[BHJ17] S. Boucksom, T. Hisamoto, and M. Jonsson, *Uniform K-stability, Duistermaat–Heckman measures and singularities of pairs*, Ann. Inst. Fourier (Grenoble) **67** (2017), no. 2, 743–841.

[Bou72] N. Bourbaki, *Commutative algebra. Chapters I–VII*, Elements of Mathematics, Hermann, Paris; Addison-Wesley, Reading, MA (1972).

[Che25] Z. Chen, *Finite generation of higher rank quasi-monomial valuations via the extended Rees algebra*, arXiv:2510.10737.

[ELS03] L. Ein, R. Lazarsfeld, and K. E. Smith, *Uniform approximation of Abhyankar valuation ideals in smooth function fields*, Amer. J. Math. **125** (2003), no. 2, 409–440.

[Eis95] D. Eisenbud, *Commutative algebra with a view toward algebraic geometry*, Graduate Texts in Mathematics **150**, Springer-Verlag, New York (1995).

[Fuj19] K. Fujita, *A valuative criterion for uniform K-stability of $\mathbb{Q}$-Fano varieties*, J. Reine Angew. Math. **751** (2019), 309–338.

[FO18] K. Fujita and Y. Odaka, *On the K-stability of Fano varieties and anticanonical divisors*, Tohoku Math. J. (2) **70** (2018), no. 4, 511–521.

[HMX14] C. D. Hacon, J. McKernan, and C. Xu, *ACC for log canonical thresholds*, Ann. of Math. (2) **180** (2014), no. 2, 523–571.

[HL23] J. Han and C. Li, *On the Yau–Tian–Donaldson conjecture for generalized Kähler–Ricci soliton equations*, Comm. Pure Appl. Math. **76** (2023), no. 9, 1793–1867.

[HL24] J. Han and C. Li, *Algebraic uniqueness of Kähler–Ricci flow limits and optimal degenerations of Fano varieties*, Geom. Topol. **28** (2024), no. 2, 539–592.

[HMQ+26] J. Han, M. Miao, L. Qi, L. Wang, and T. Zhang, *Stable degenerations of log Fano fibration germs*, arXiv:2603.09269.

[Har77] R. Hartshorne, *Algebraic geometry*, Graduate Texts in Mathematics, No. 52, Springer-Verlag, New York–Heidelberg (1977).

[JM12] M. Jonsson and M. Mustaţă, *Valuations and asymptotic invariants for sequences of ideals*, Ann. Inst. Fourier (Grenoble) **62** (2012), no. 6, 2145–2209.

[Kim26] D. Kim, *On the existence of minimizer on a log Fano cone singularity*, arXiv:2608.23019.

[KL26] D. Kim and D.-W. Lee, *On the quasi-monomiality of the α- and δ-invariants*, arXiv:2604.18465.

[Kol07] J. Kollár, *Lectures on resolution of singularities*, Annals of Mathematics Studies **166**, Princeton University Press, Princeton, NJ (2007).

[KM98] J. Kollár and S. Mori, *Birational geometry of algebraic varieties*, Cambridge Tracts in Math. **134**, Cambridge Univ. Press, Cambridge (1998).

[Laz04] R. Lazarsfeld, *Positivity in algebraic geometry. I. Classical setting: line bundles and linear series*, Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge **48**, Springer-Verlag, Berlin (2004).

[Li17] C. Li, *K-semistability is equivariant volume minimization*, Duke Math. J. **166** (2017), no. 16, 3147–3218.

[LX18] C. Li and C. Xu, *Stability of valuations: higher rational rank*, Peking Math. J. **1** (2018), no. 1, 1–79.

[LXZ22] Y. Liu, C. Xu, and Z. Zhuang, *Finite generation for valuations computing stability thresholds and applications to K-stability*, Ann. of Math. (2) **196** (2022), no. 2, 507–566.

[MFK94] D. Mumford, J. Fogarty, and F. Kirwan, *Geometric invariant theory*, 3rd ed., Ergebnisse der Mathematik und ihrer Grenzgebiete (2) **34**, Springer-Verlag, Berlin (1994).

[PS09] Yu. G. Prokhorov and V. V. Shokurov, *Towards the second main theorem on complements*, J. Algebraic Geom. **18** (2009), no. 1, 151–199.

[Roc70] R. T. Rockafellar, *Convex analysis*, Princeton Mathematical Series, No. 28, Princeton University Press, Princeton, NJ (1970).

[RTZ21] Y. A. Rubinstein, G. Tian, and K. Zhang, *Basis divisors and balanced metrics*, J. Reine Angew. Math. **778** (2021), 171–218.

[Sho92] V. V. Shokurov, *Threefold log flips*, with an appendix in English by Y. Kawamata, Izv. Ross. Akad. Nauk Ser. Mat. **56** (1992), no. 1, 105–203.

[Wan25] L. Wang, *Generalized optimal degenerations of Fano varieties*, Peking Math. J. (2025), published online, doi:10.1007/s42543-025-00112-3.

[Xu20] C. Xu, *A minimizing valuation is quasi-monomial*, Ann. of Math. (2) **191** (2020), no. 3, 1003–1030.

[Xu21] C. Xu, *Towards finite generation of higher rational rank valuations*, Mat. Sb. **212** (2021), no. 3, 157–174; English transl. in Sb. Math. **212** (2021), 416–432.

[XZ20] C. Xu and Z. Zhuang, *On positivity of the CM line bundle on K-moduli spaces*, Ann. of Math. (2) **192** (2020), no. 3, 1005–1068.

[XZ25] C. Xu and Z. Zhuang, *Stable degenerations of singularities*, J. Amer. Math. Soc. **38** (2025), no. 3, 585–626.

[XZ26] C. Xu and Z. Zhuang, *Open problems in K-stability of Fano varieties*, arXiv:2601.15576.

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